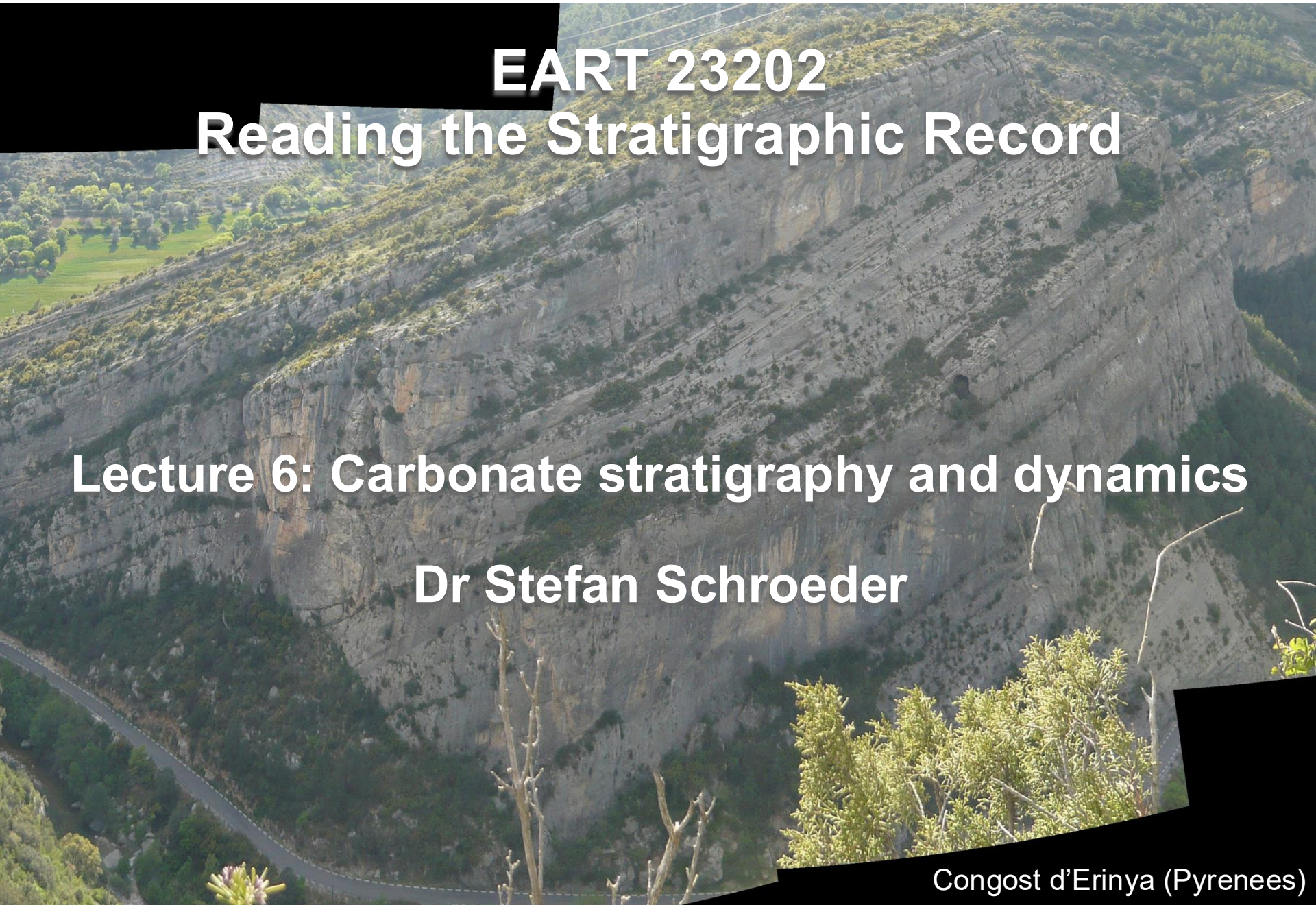
EART 23202
Reading the Stratigraphic Record

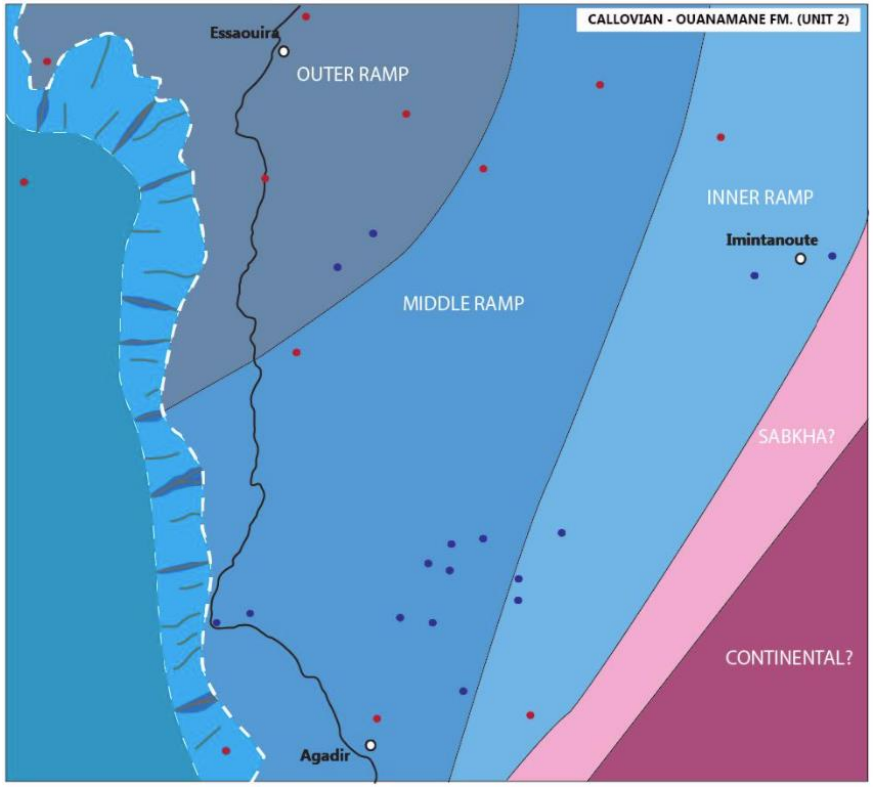


Lecture 6: Carbonate stratigraphy and dynamics

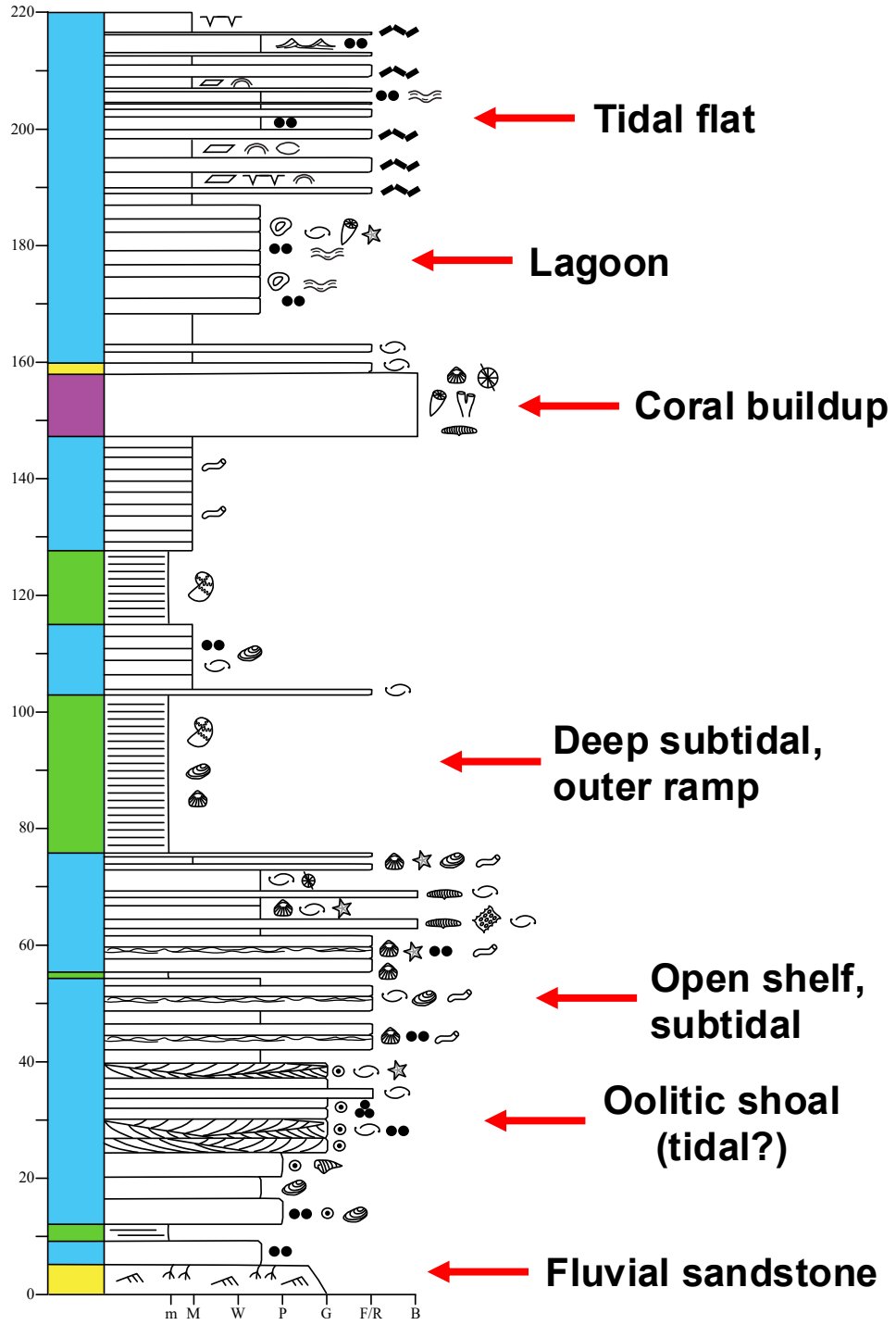
Dr Stefan Schroeder

Congost d'Erinya (Pyrenees)

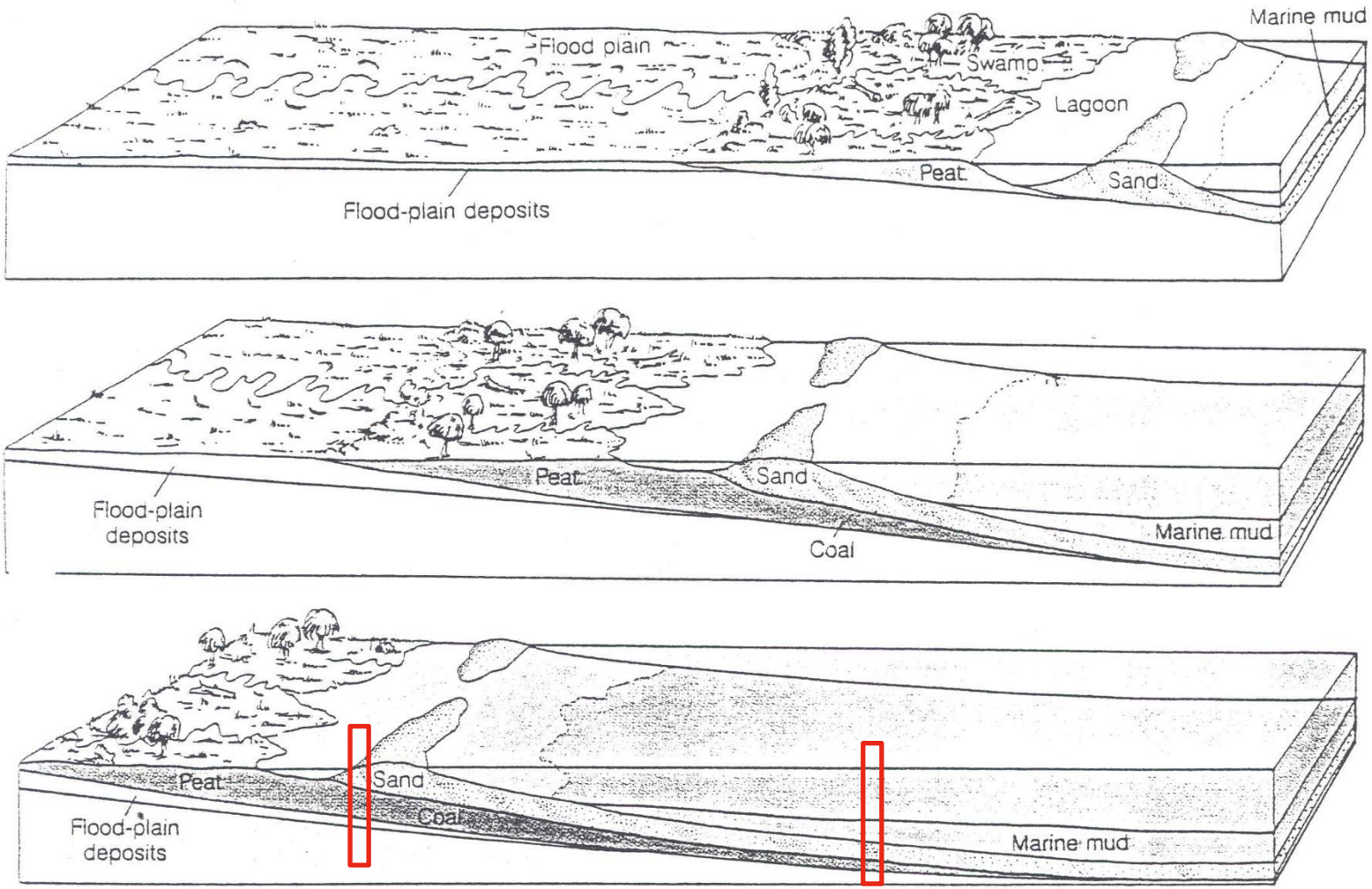
Carbonate dynamics
Paleoenvironments in space & time



Duval-Arnould 2019



Carbonate dynamics
Paleoenvironments in space & time

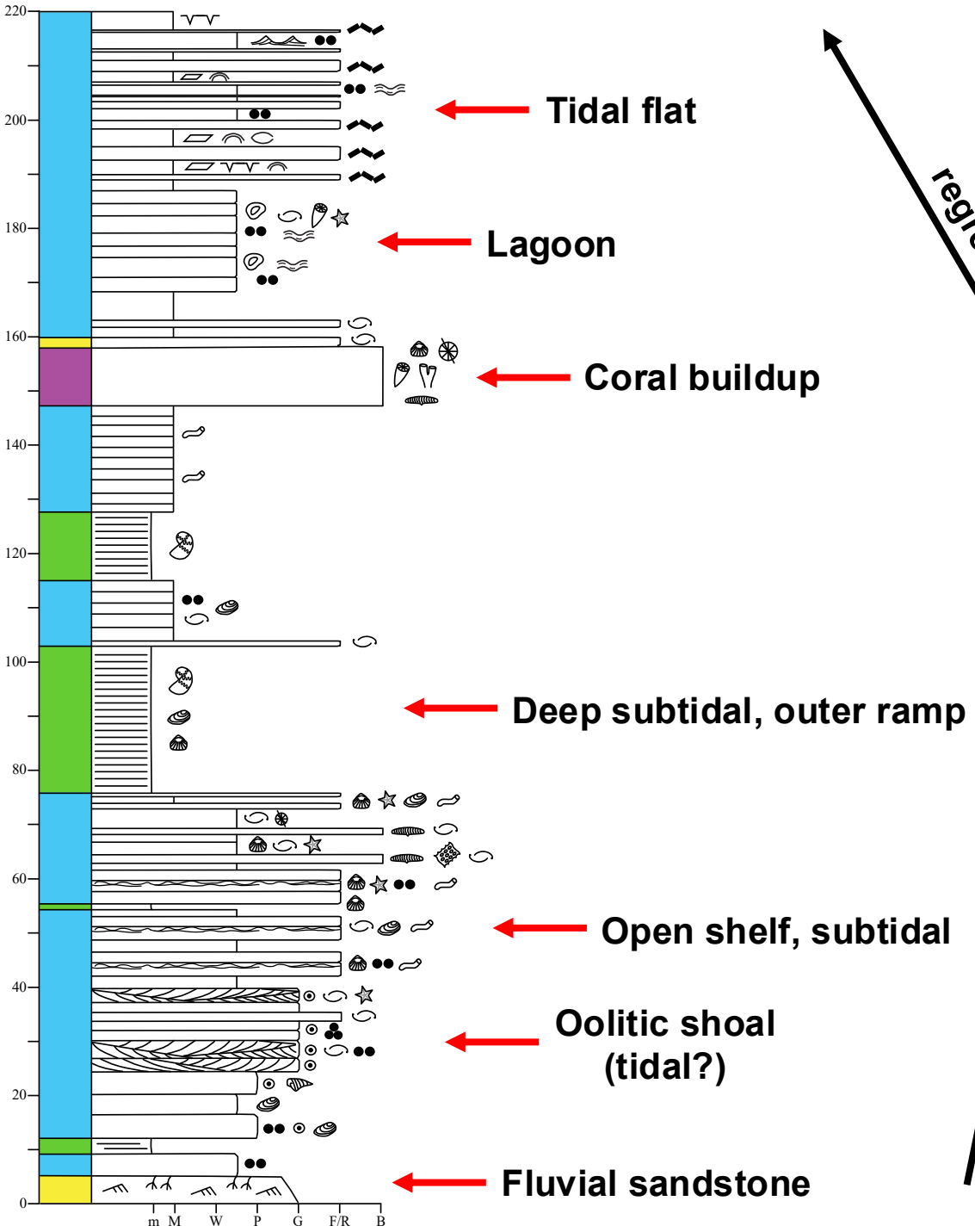


Walther's Law: "Facies occurring in a conformable vertical sequence were formed in laterally adjacent environments."

- Lateral migration of facies belts due to extrinsic or intrinsic forcing leads to vertical superposition of facies with time

Carbonate dynamics

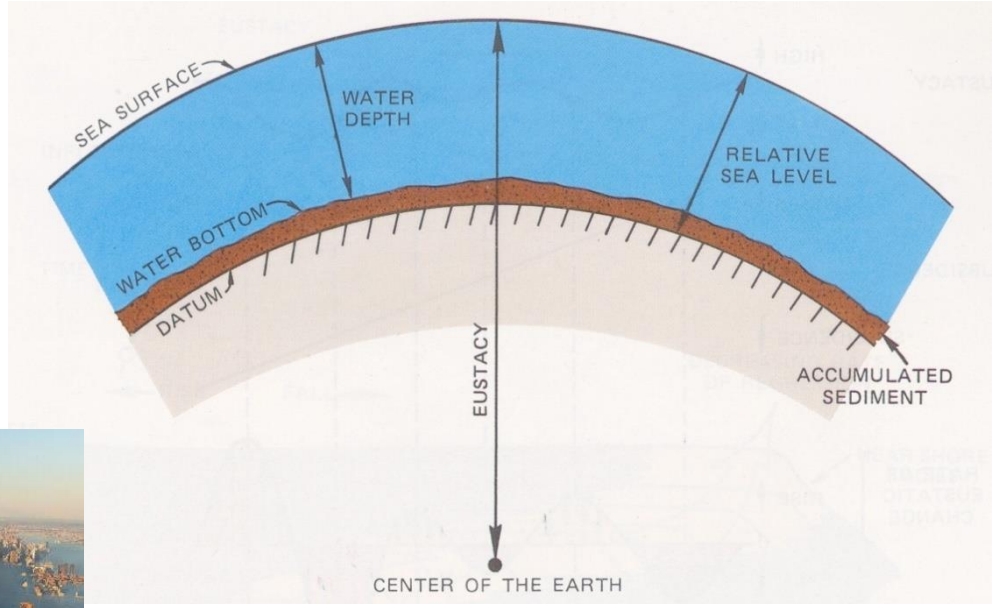
Paleoenvironments in space & time



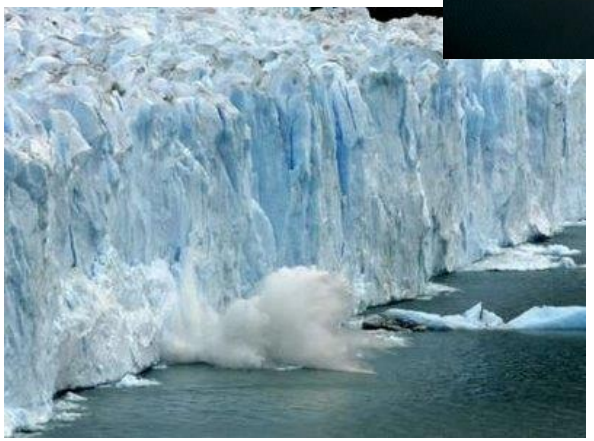
Carbonates and sea level

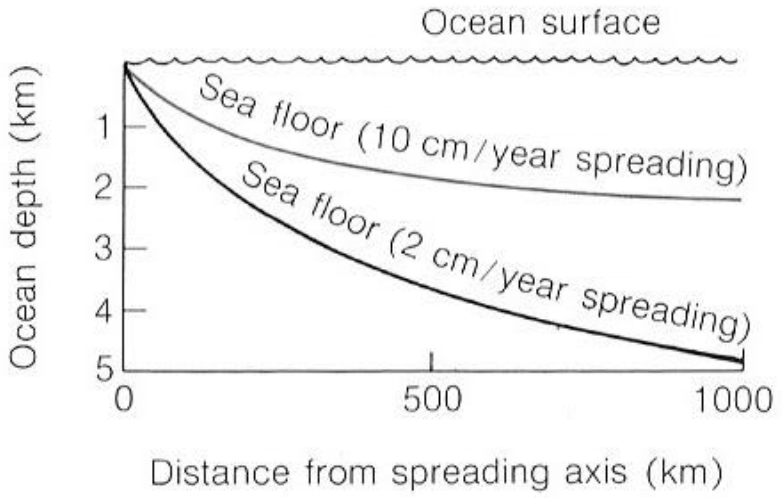
Carbonate response to sea level change
Eustasy

- **Eustatic sea level change is controlled at long time scales >10^6 years) by:**
  - **Volume of water in oceans – especially driven by glacial ice volumes**
  - **Volume of ocean basins – related to spreading rates/plate tectonics**



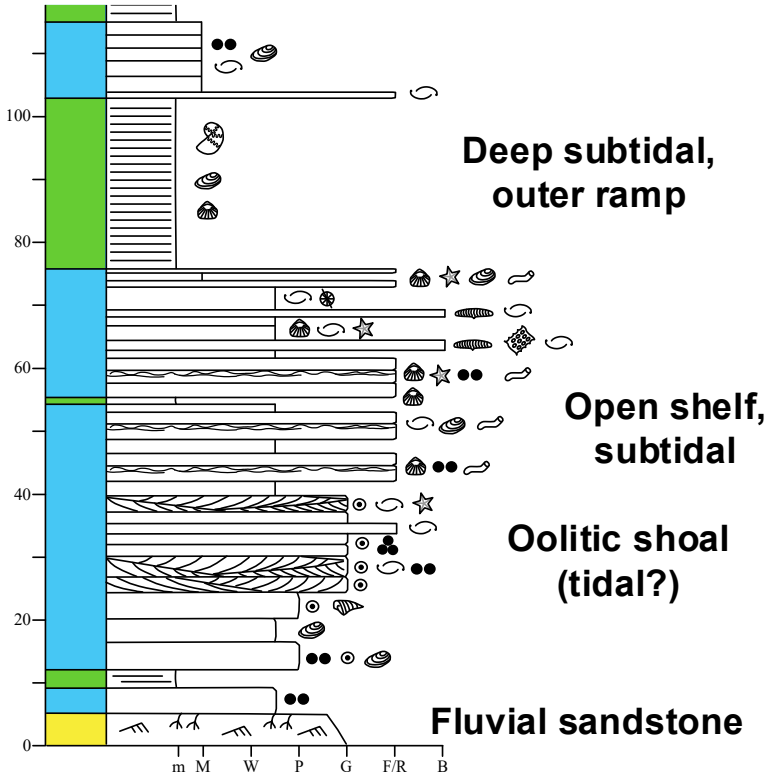


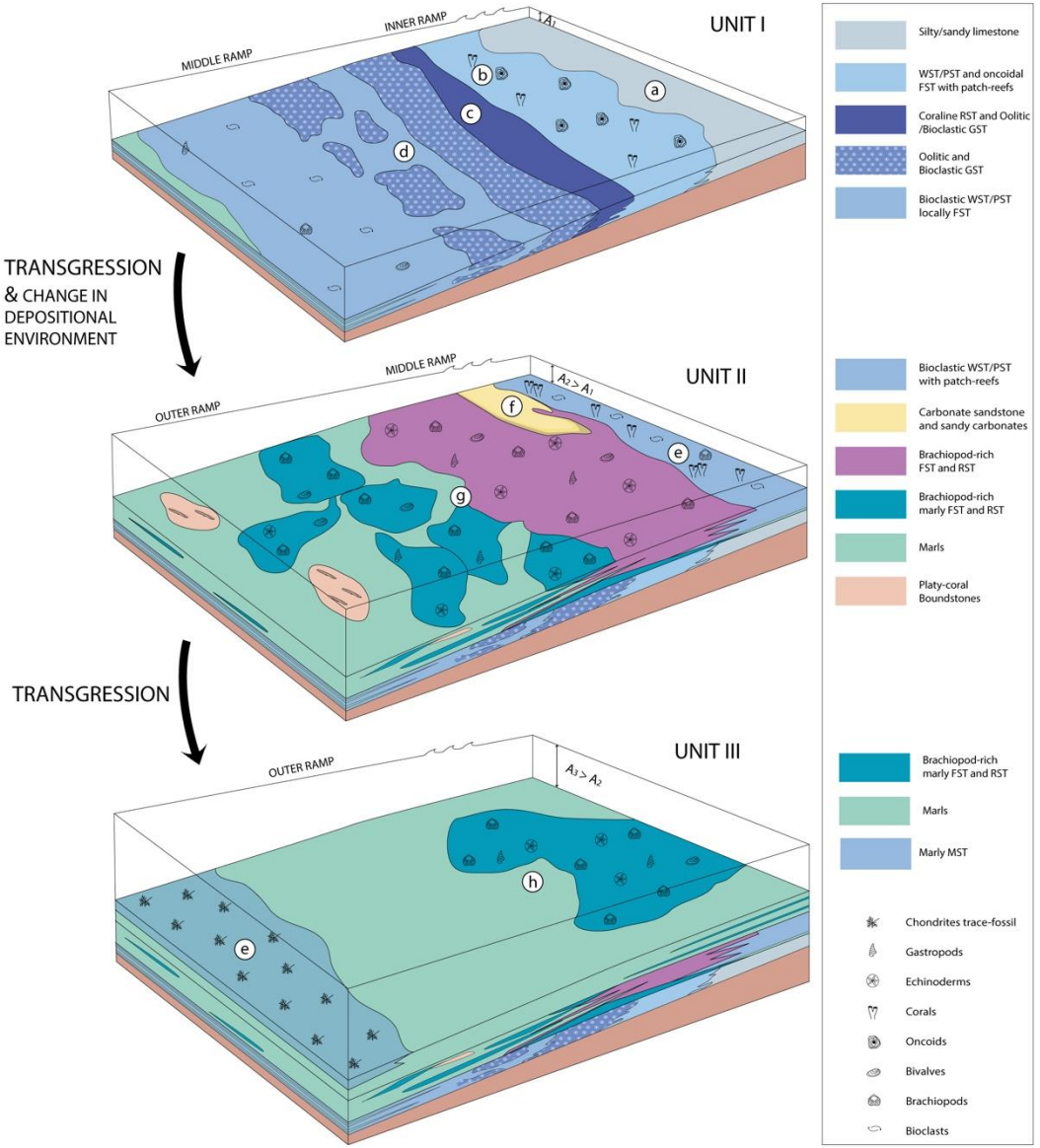




Press & Siever (1986)

Carbonate response to sea level change
Transgression



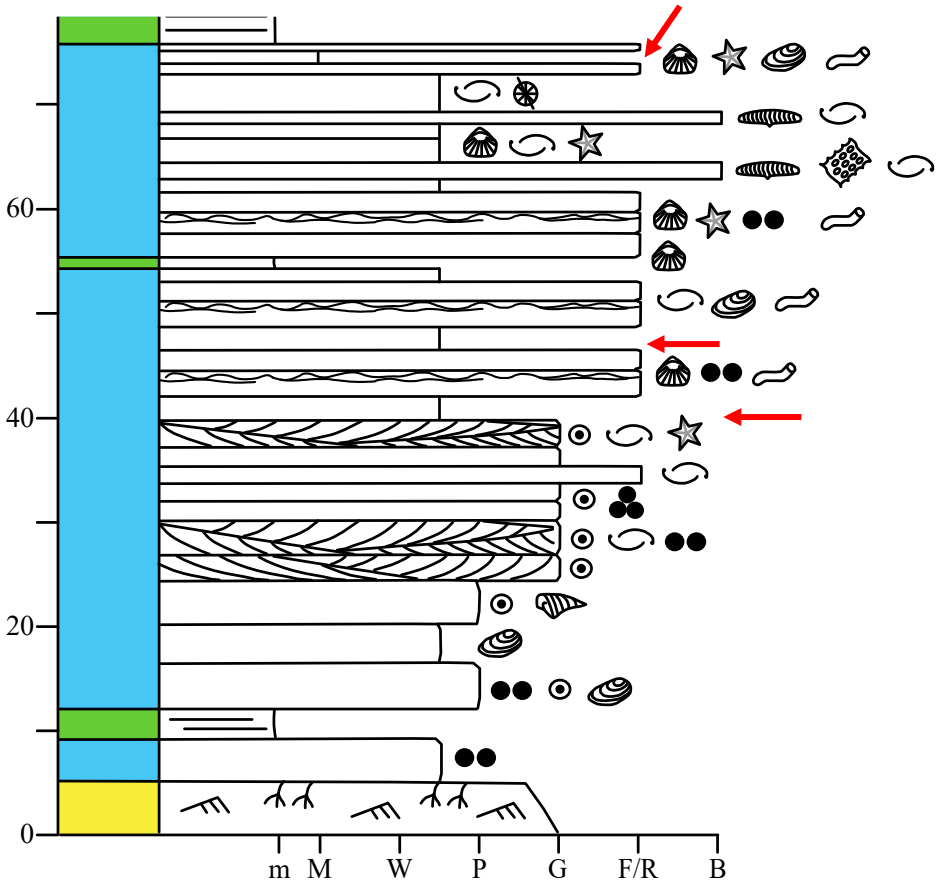


Exercise 7



Exercise 7





Several bed tops in the open shelf unit show additional structures.
Try to identify these from these images, and refer to the hand specimens in class.

Exercise 7

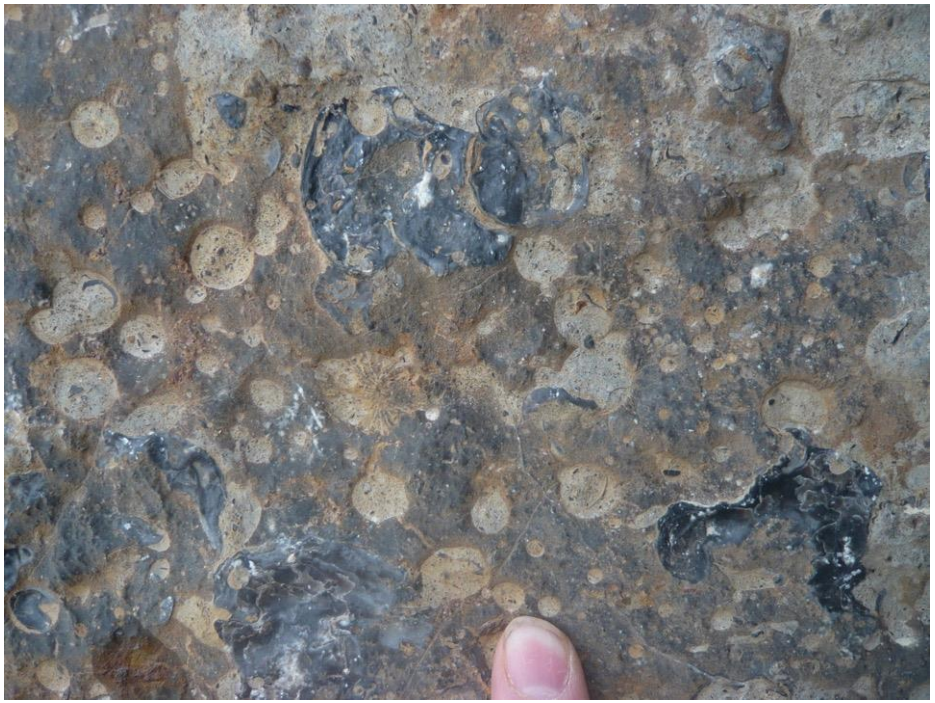


Picture of a bedding plane. What allochems and sedimentary structures can you identify on this bedding plane?

Exercise 7



What allochems and sedimentary structures can you identify on this bedding plane?



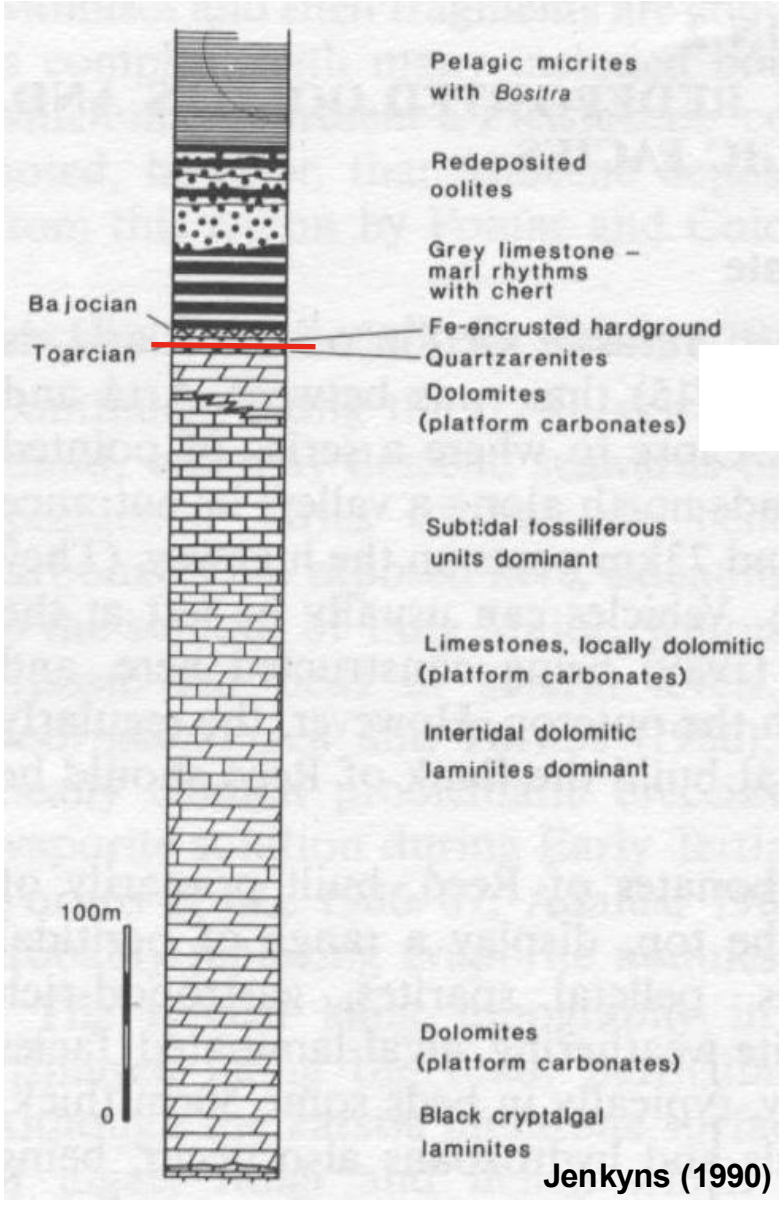
Exercise 7

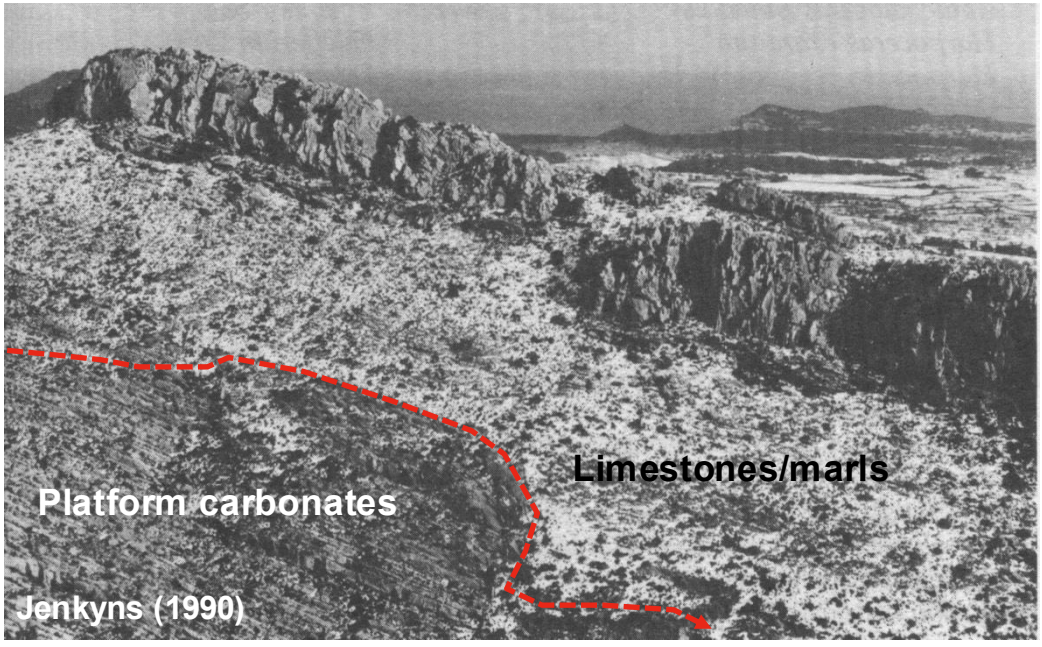


What are the ring-like structures on this bedding plane?

What do these structures and allochems tell us about sedimentary processes affecting bedding planes?
What do these processes mean for relative sea level?

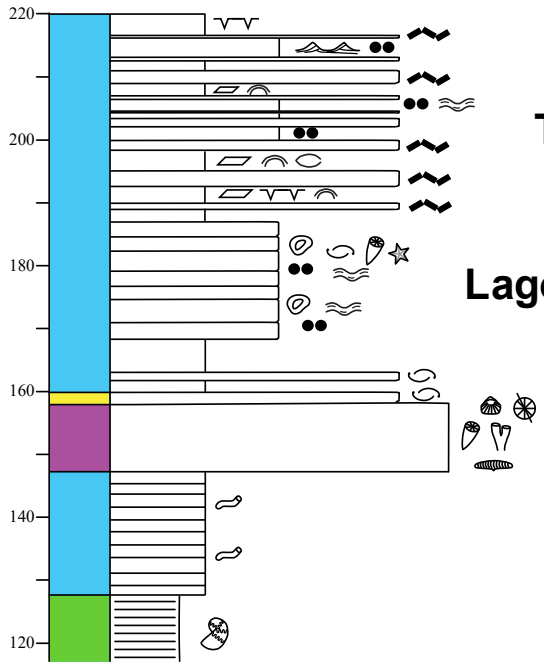
Carbonate response to sea level change
Transgression







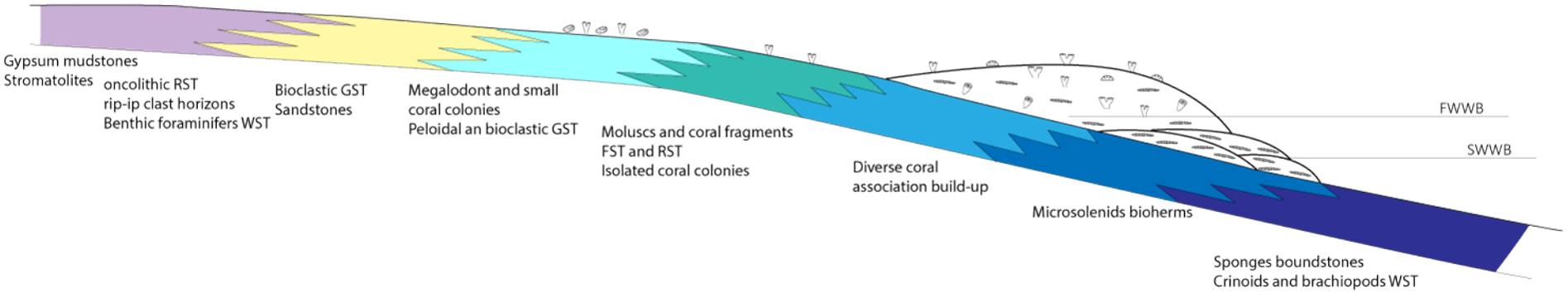
Carbonate response to sea level change
Regression



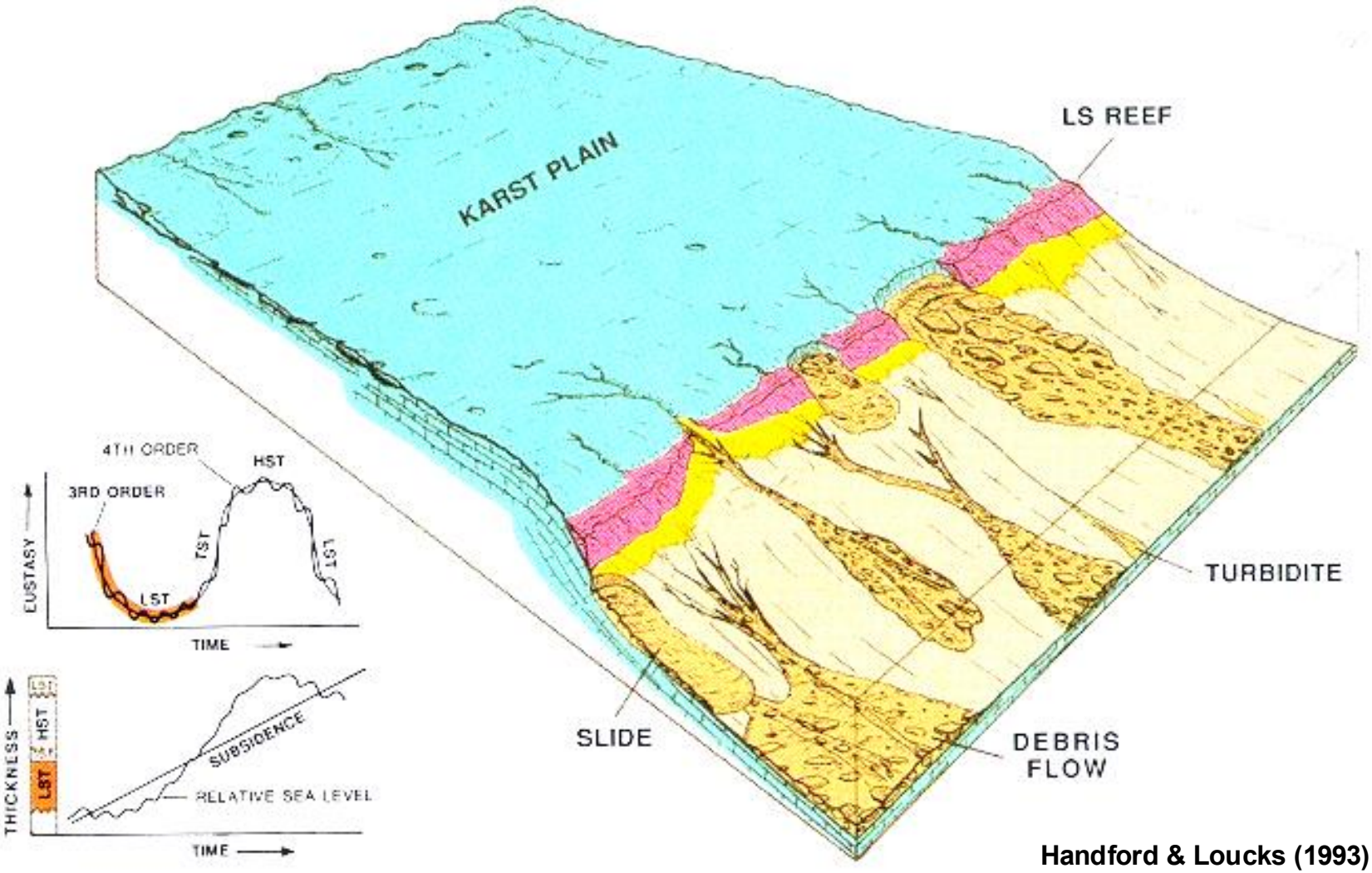
Tidal flat

Lagoon

Coral buildup



Carbonate response to sea level change
Sea-level fall



Handford & Loucks (1993)

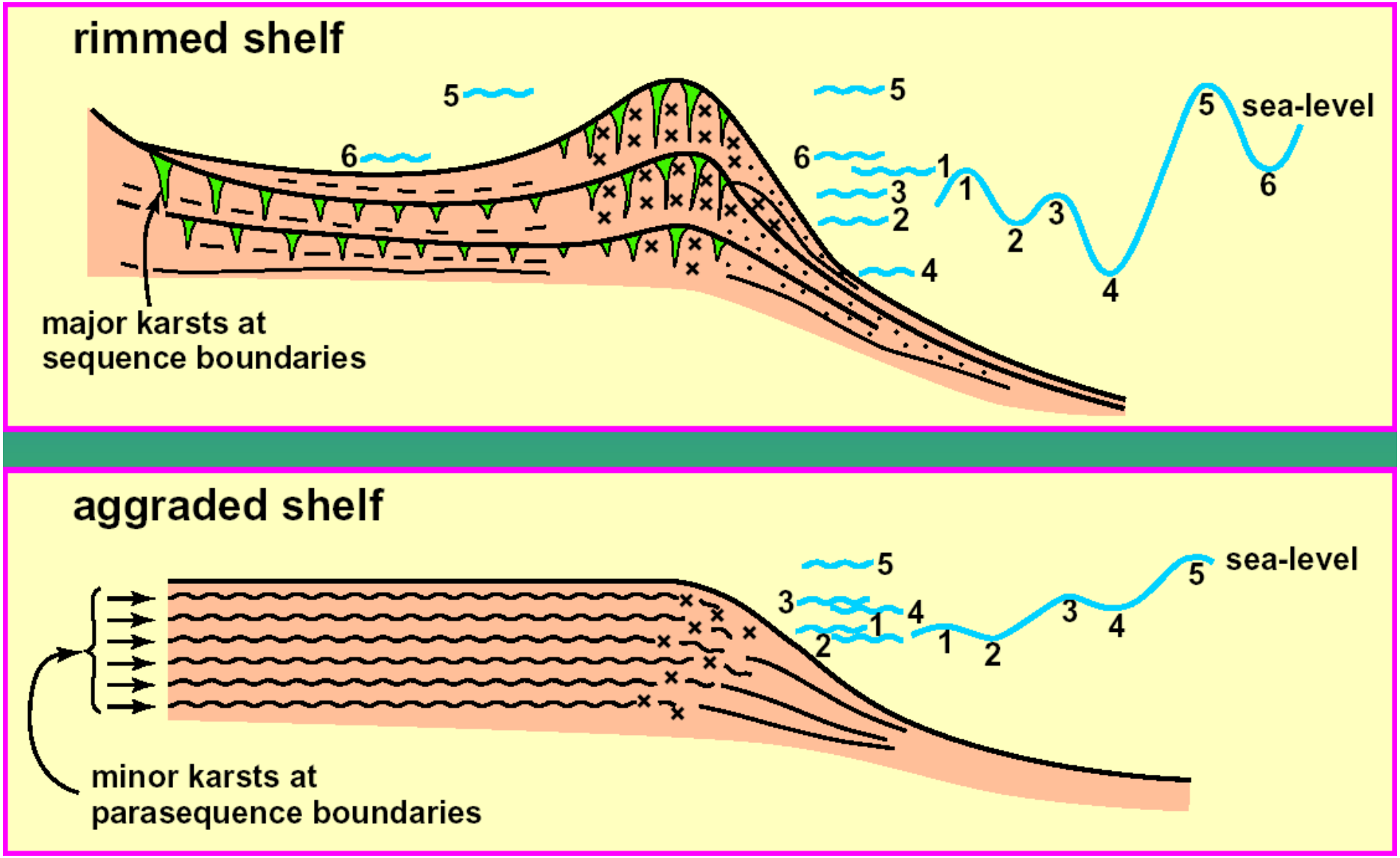
Carbonate response to sea level change
Sea-level fall







Carbonate response to sea level change
Karst



Tucker (1993)

Peritidal shallowing-up successions

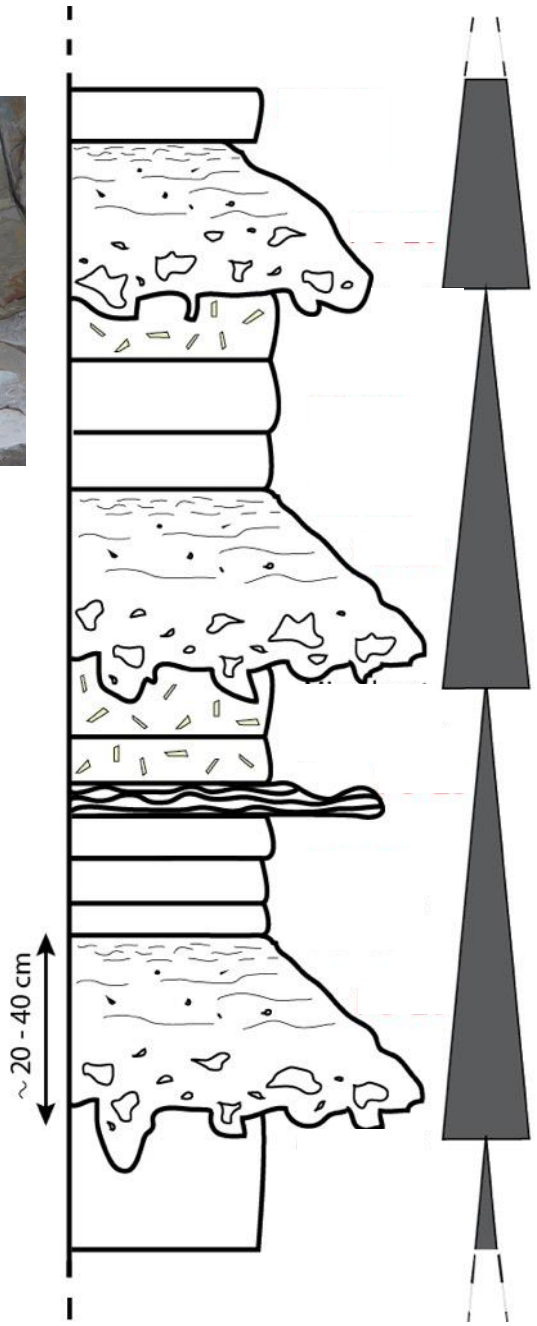
Peritidal carbonates
Shallowing-up successions





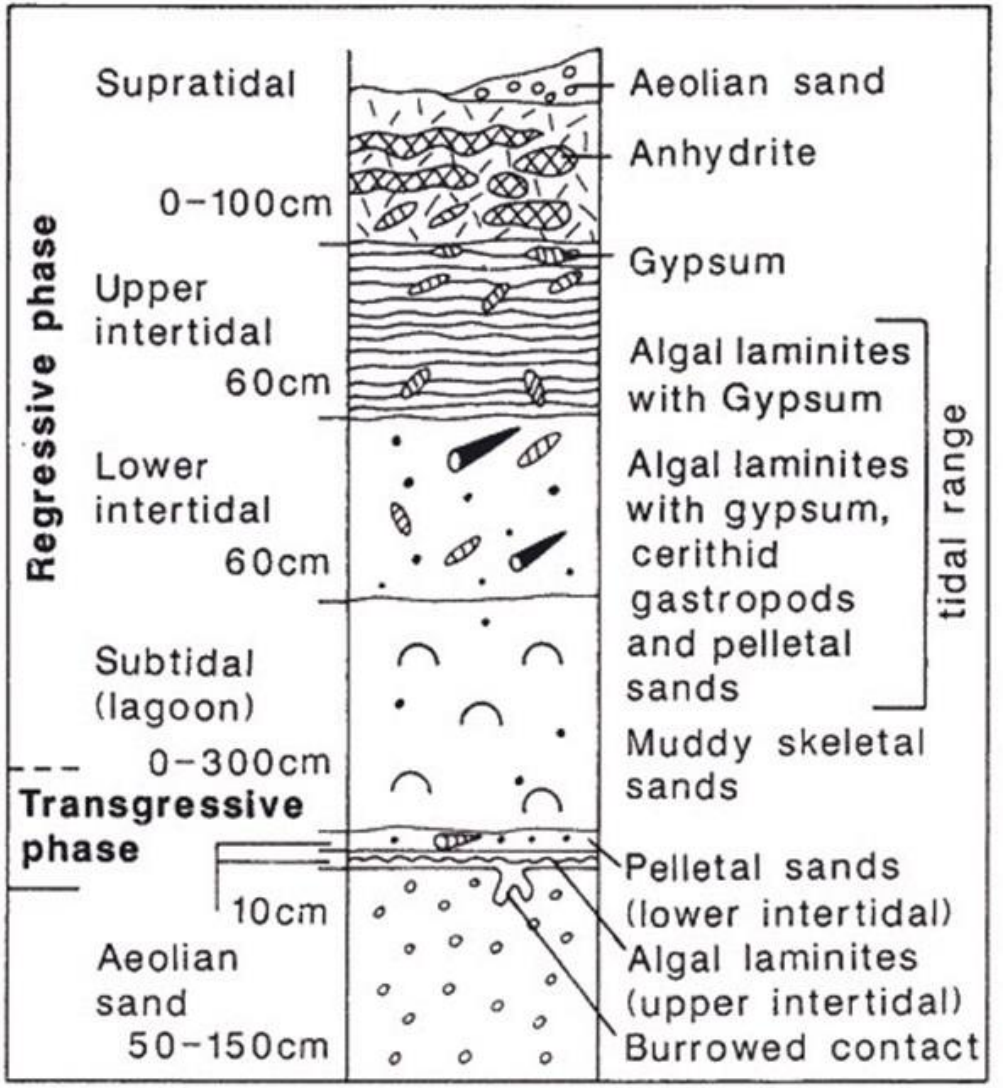






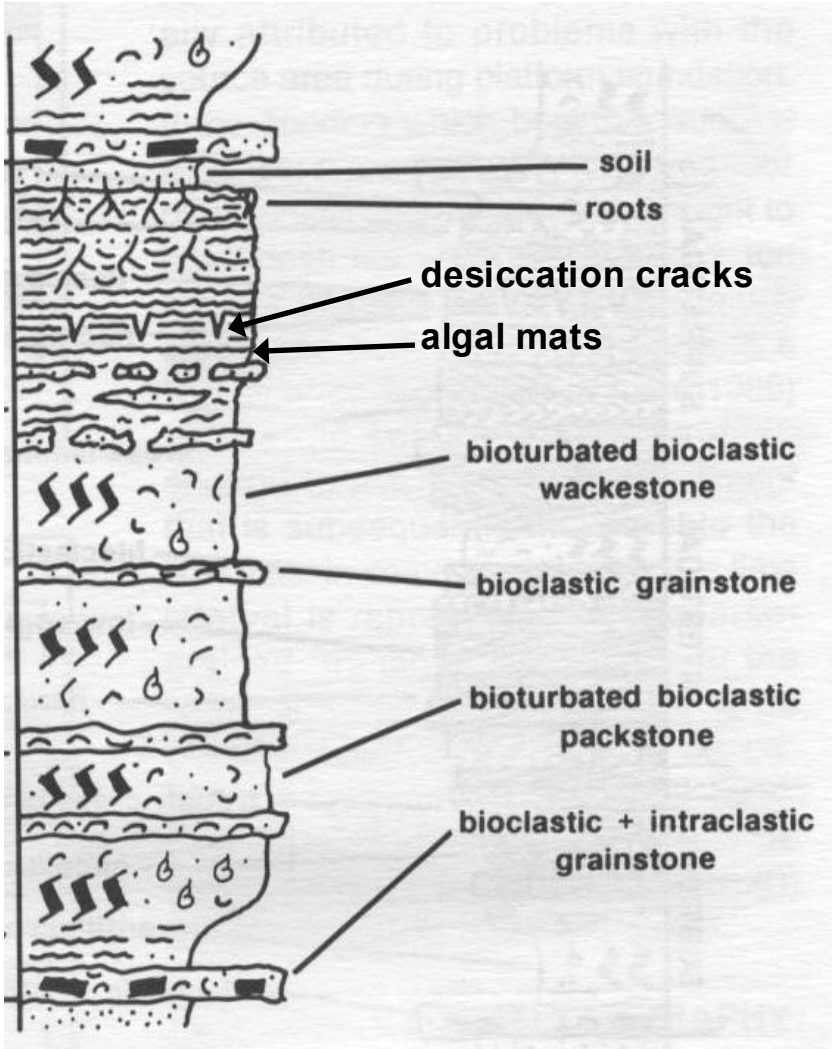
Peritidal carbonates
Shallowing-up successions

Arid climate



Tucker (1990)

Humid climate

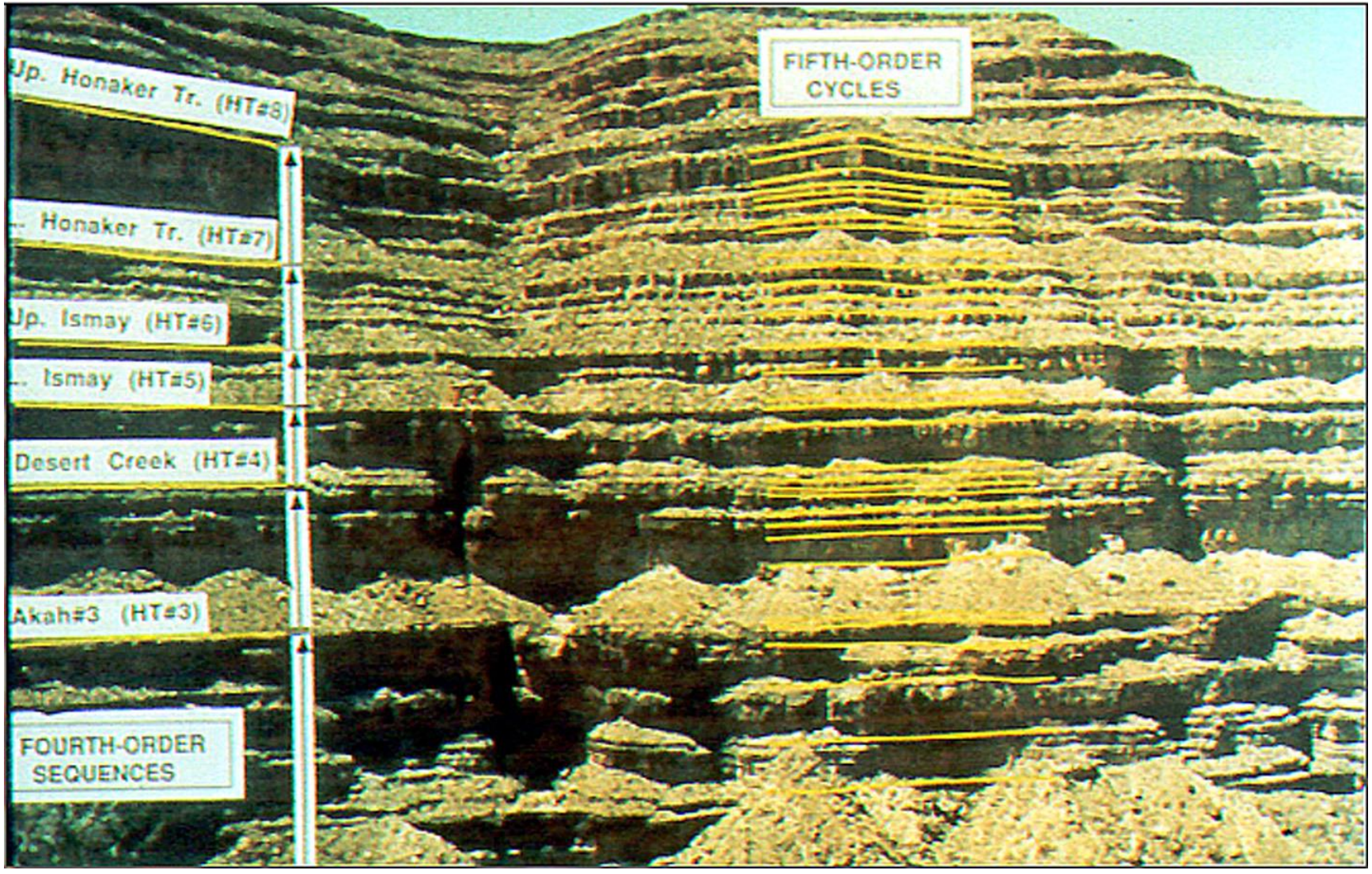


Walker (1984)

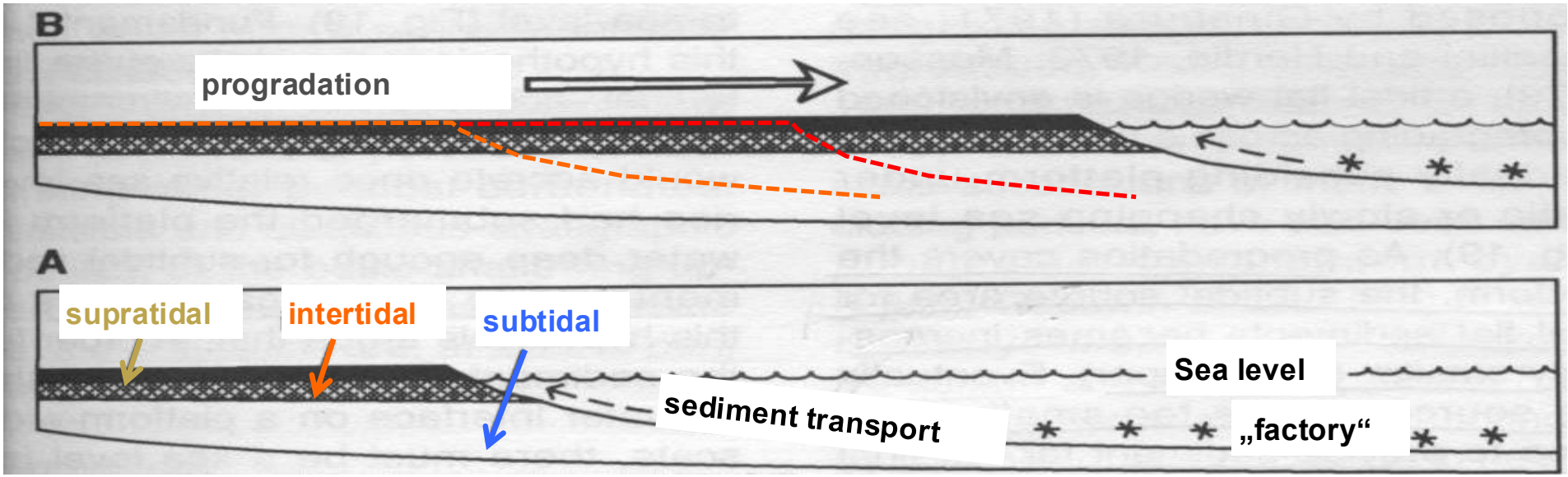
Peritidal carbonates
Shallowing-up successions

- Most modern and ancient carbonates have stacked vertical successions of increasingly shallow water facies
- Evidence of seaward facies migration
- Controls may be related to sedimentary processes and/or eustatic sea level fluctuations

Subtidal to exposure-capped parasequences - Paradox Basin, Utah (Exxon Mobil outcrop atlas)



Peritidal carbonates
Shallowing-up successions



Walker (1984)

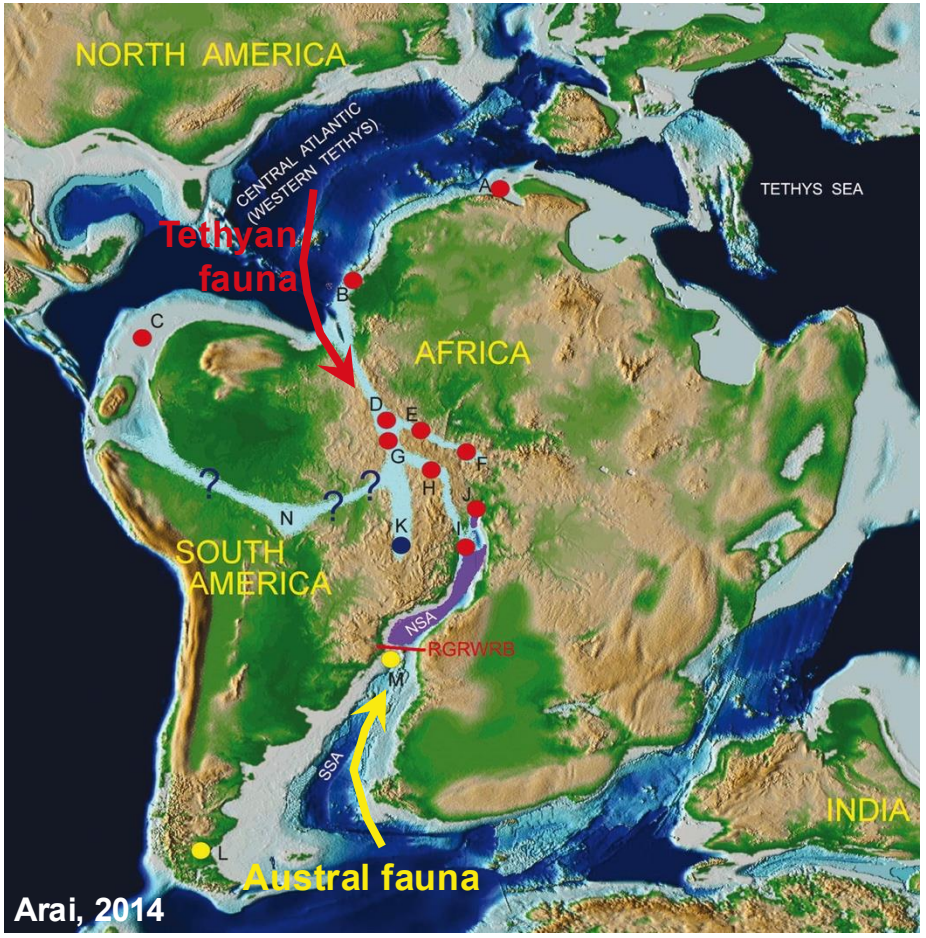
Carbonates as paleoclimate and paleobiology records

Carbonates record evolution of life

Archean (2.5 Ga old) stromatolites, South Africa



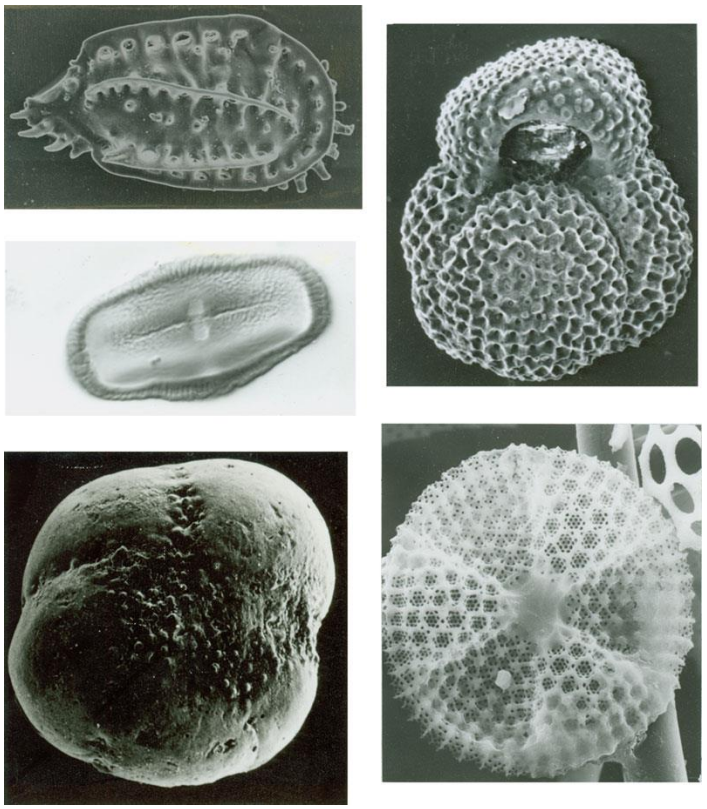




Early Cretaceous: reconstruction of southern Atlantic seaways using fauna

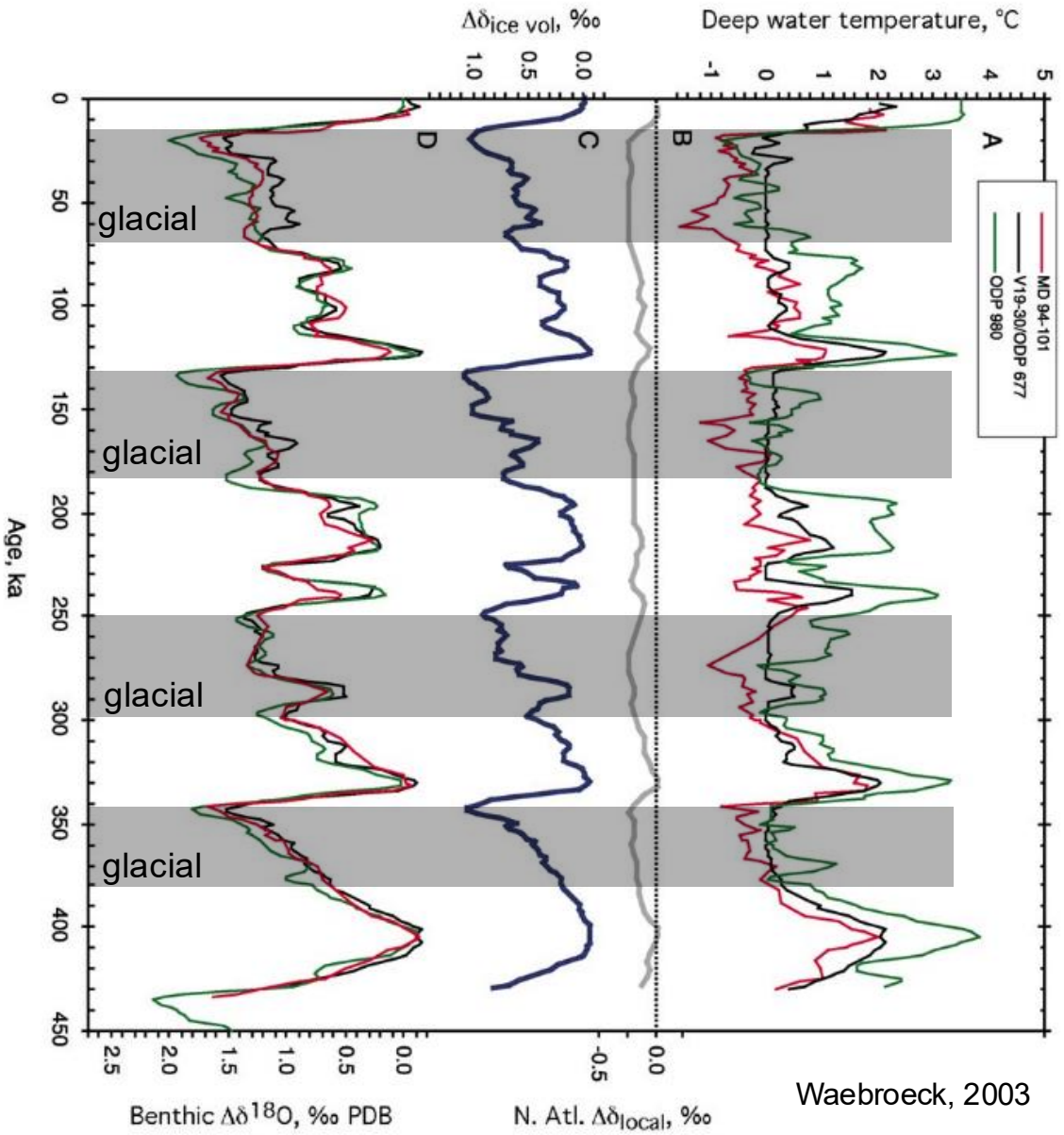
Jurassic ammonites and other skeletal fauna (~160 Ma), Spain

Carbonates record past climates and environments



© USGS

Record of Quaternary ocean temperature in carbonates/fossil shells



Waebroeck, 2003

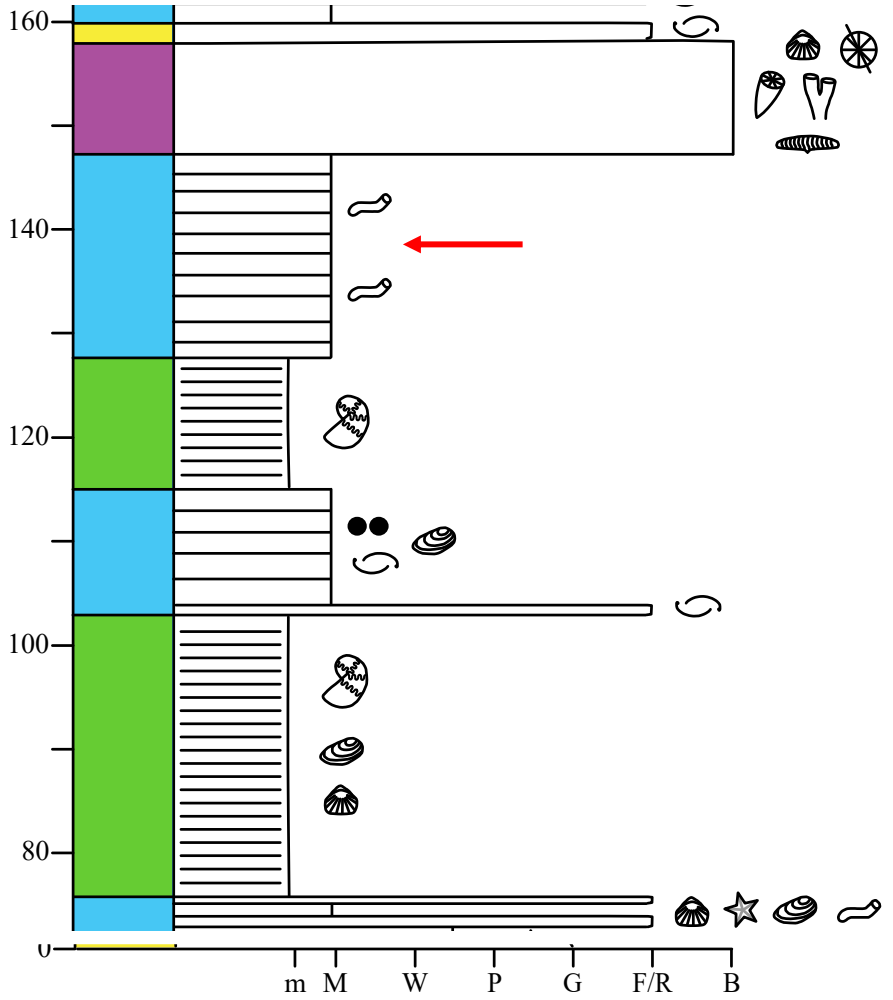
Exercise 8



Exercise 8

Unit 3a: Marl member





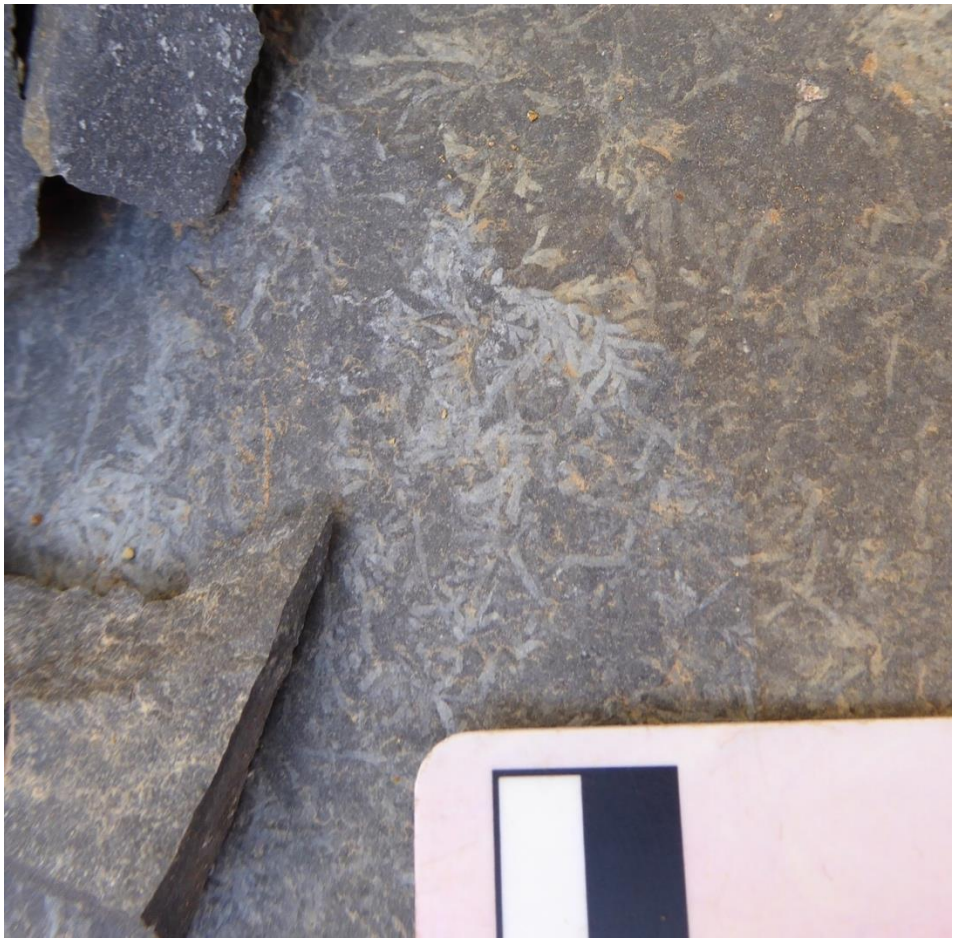
Towards the top of the marl member, limestone beds become dark grey and have a distinct kerogen smell when hitting with a hammer.

What gives these limestones the dark color?

Exercise 8

Unit 3a: Marl member



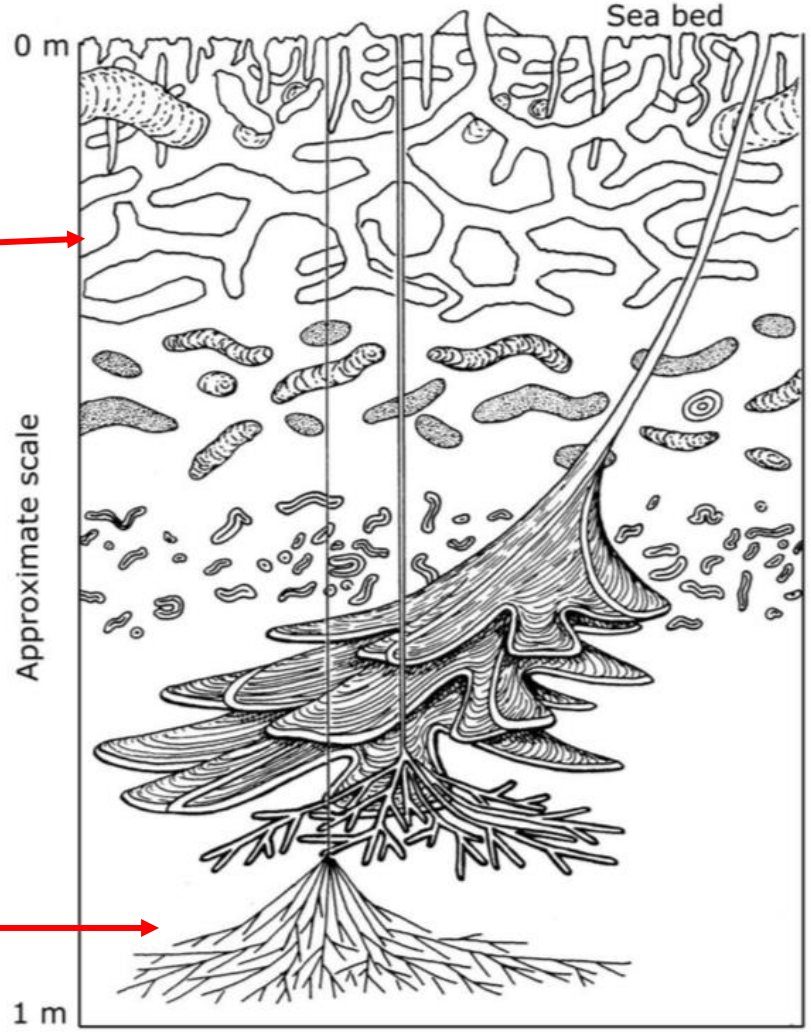


Limestones are devoid of fossils, but bedding surfaces contain this structure – what is it, and what does it tell us about the environment of deposition?

Exercise 8







Tiers

Shafts, three sizes of ?*Planolites*, large *Taenidium*

Thalassinoides

Dark *Planolites* and *Taenidium*

Anconichnus

Dark *Zoophycos*

White *Zoophycos*

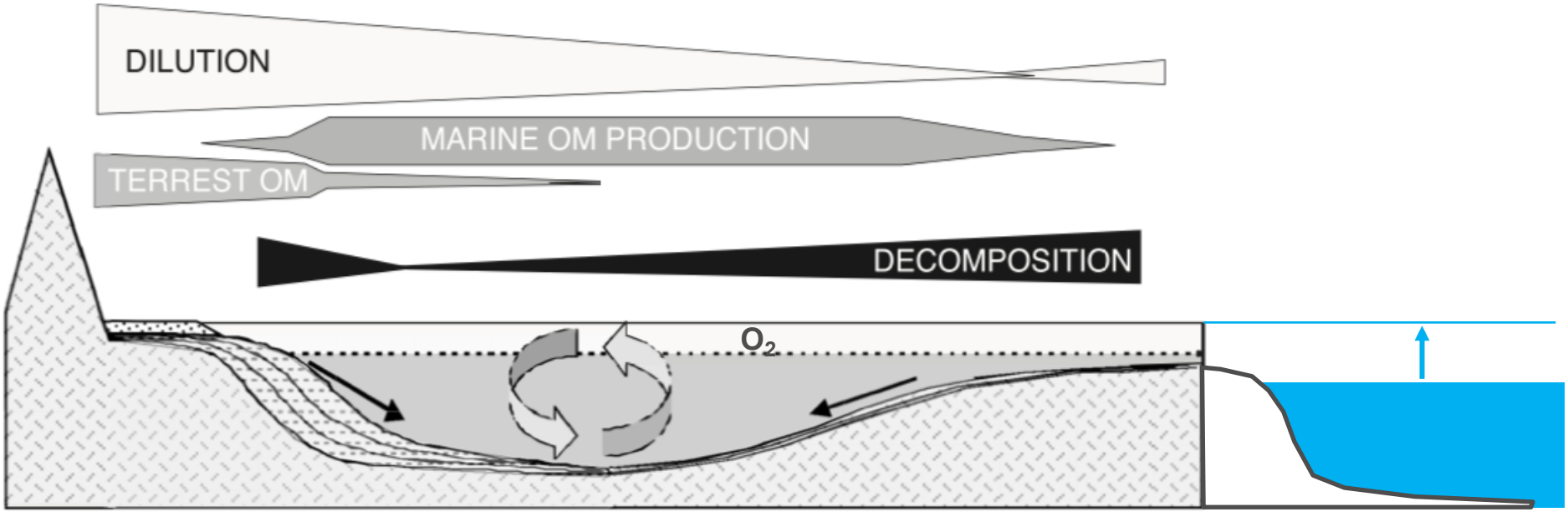
Large *Chondrites*

Medium-sized *Chondrites*

Small *Chondrites*

Lauridsen (2014)

Carbonates record past climates and environments
Ocean chemistry



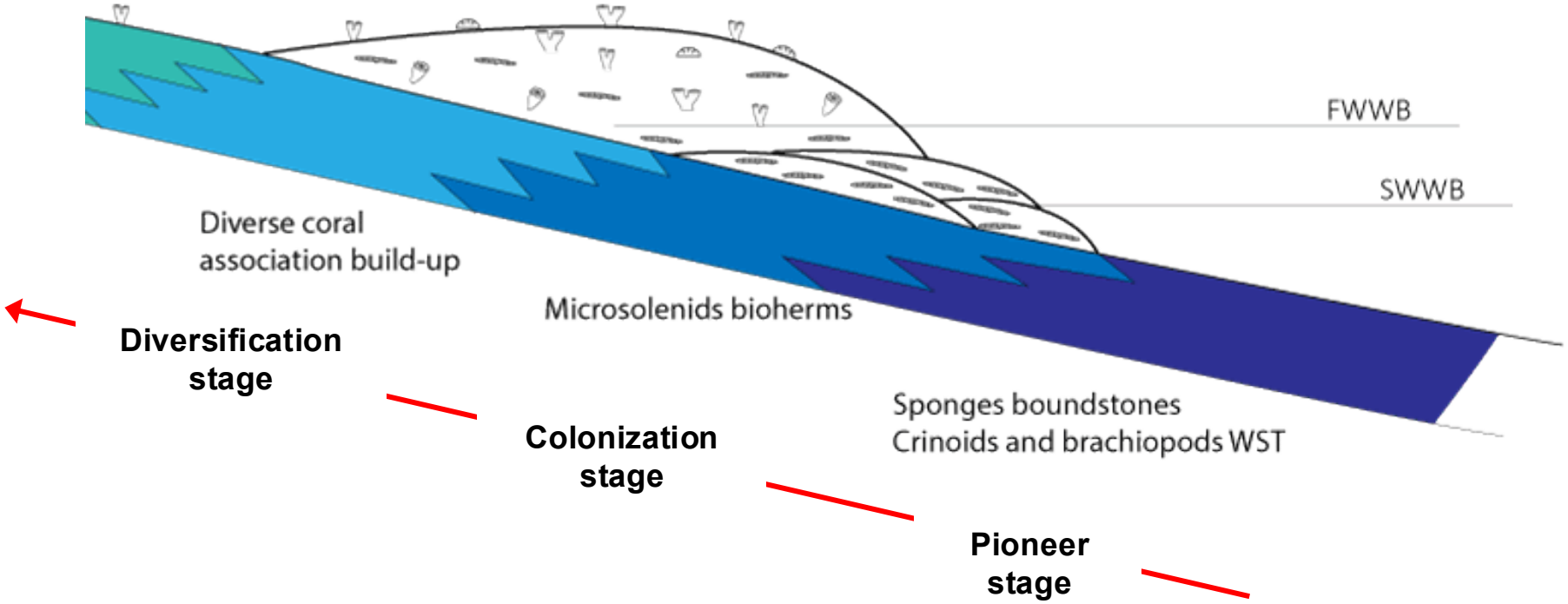
Arthur 2004

Transgressions:

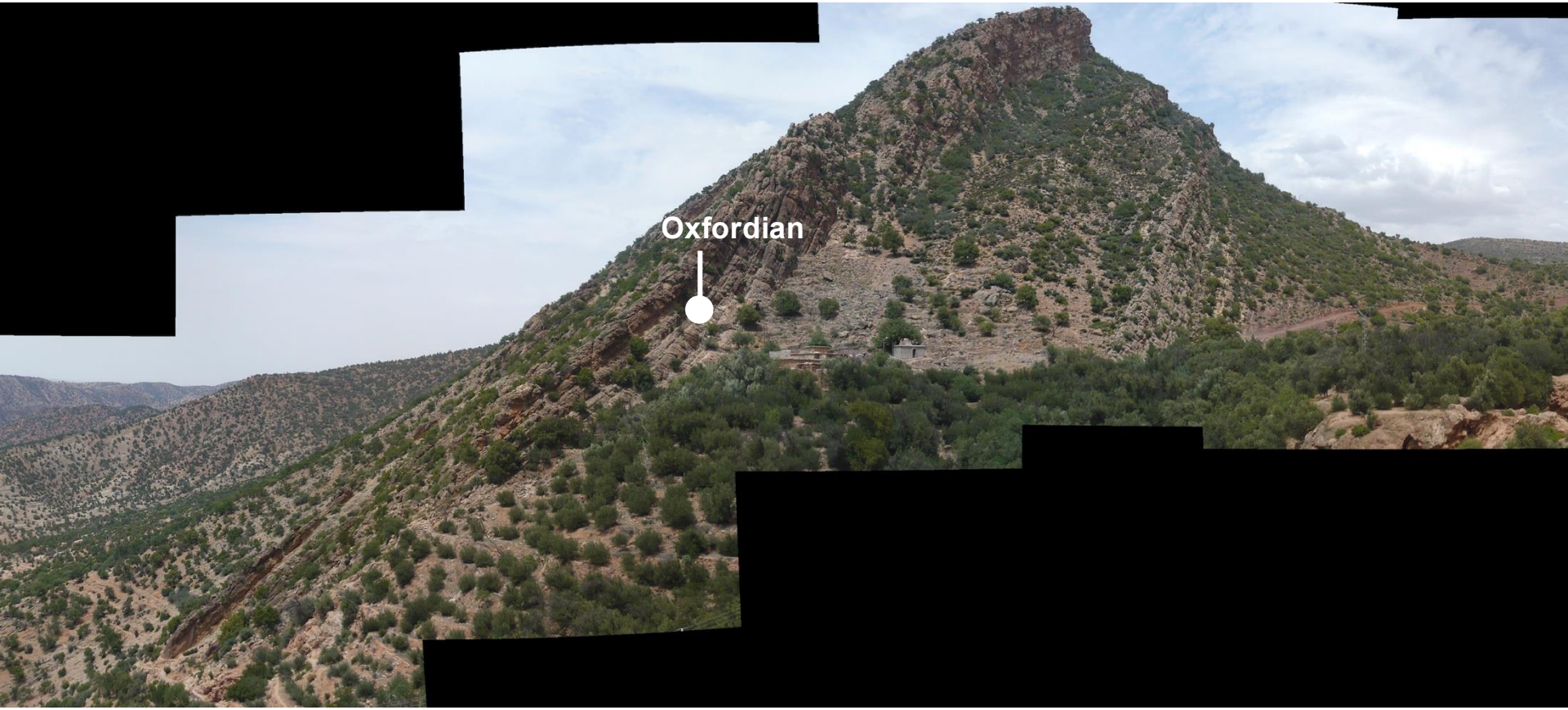
- **Advect deeper-water nutrients to shelf: increased productivity**
- **Cause retention of nutrients in shallow seas: increased productivity**
- **Promote stratification and dys/anoxia: increased preservation**
- **Cause retention of clastics nearshore: less dilution**

Carbonates record past climates and environments
Reef ecology



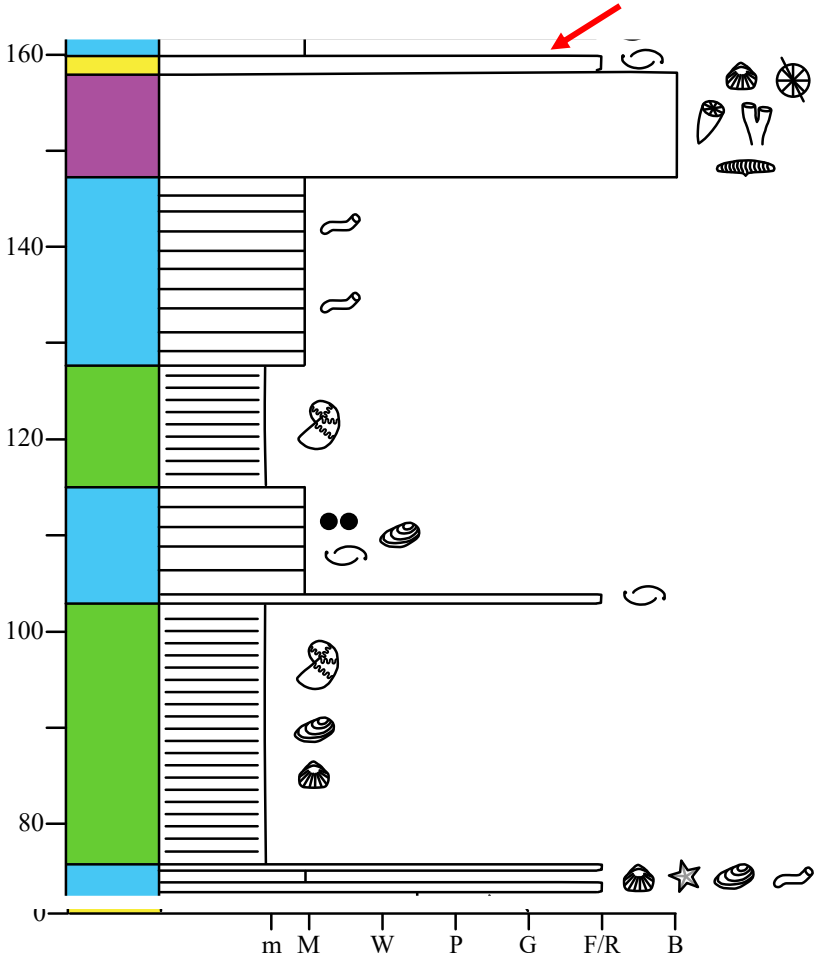


Exercise 9



Exercise 9





The buildup is overlain by a different lithology.
What are its components and sedimentary structures?

Exercise 9



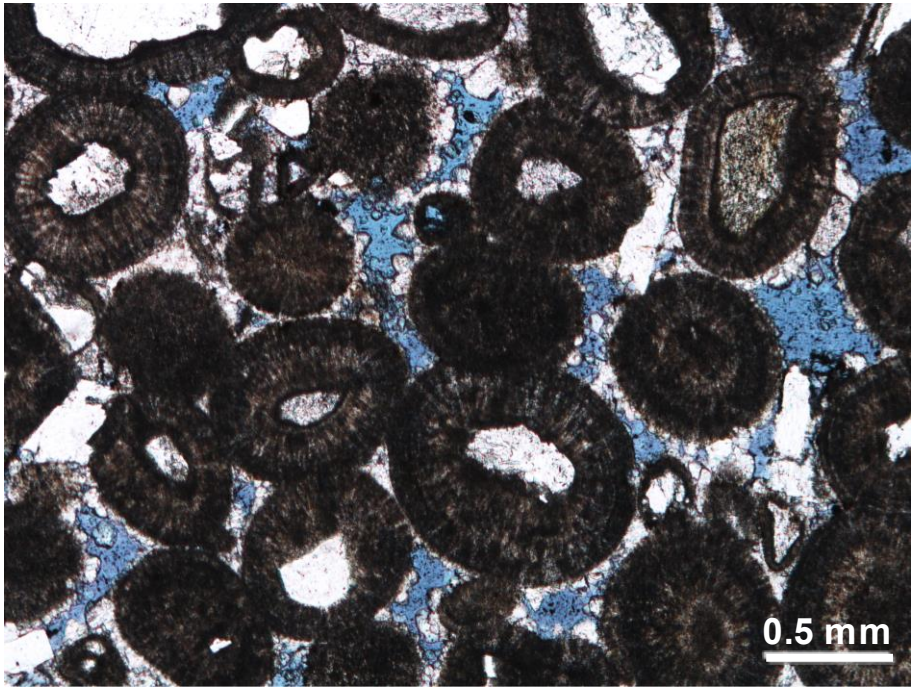


What do these observations imply for sedimentary processes and the overall environmental interpretation?

Carbonates record past climates and environments
Clastic input

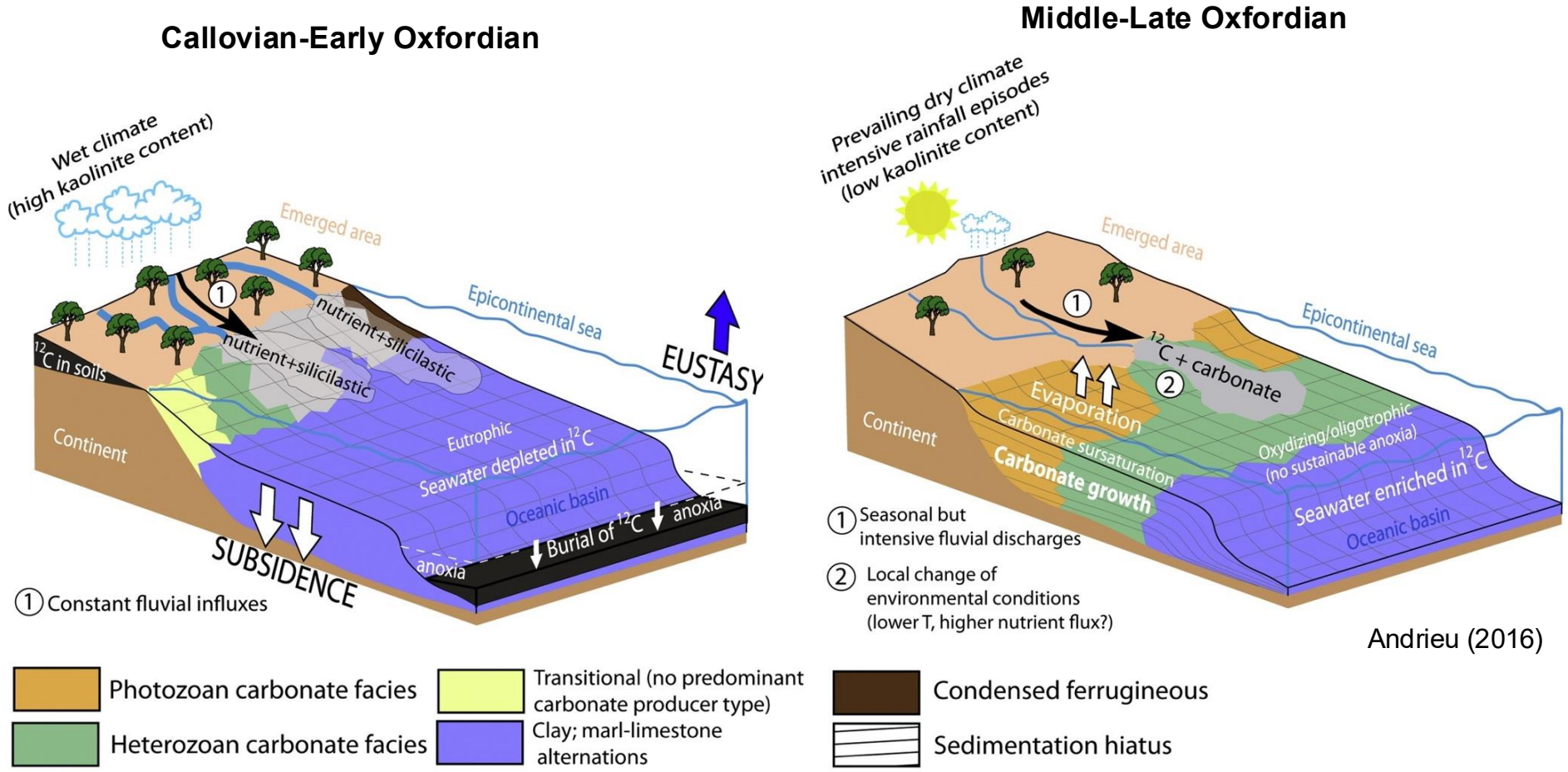
- input of clastic material inhibits carbonate production
- suspended mud increases turbidity and decreases light levels ('clastic poisoning')
- mixed carbonate-clastic deposition can occur as a result of delta lobe 'switching'





Ooid-coated quartz & feldspar, Cretaceous, Angola

Carbonates record past climates and environments
Reef ecology

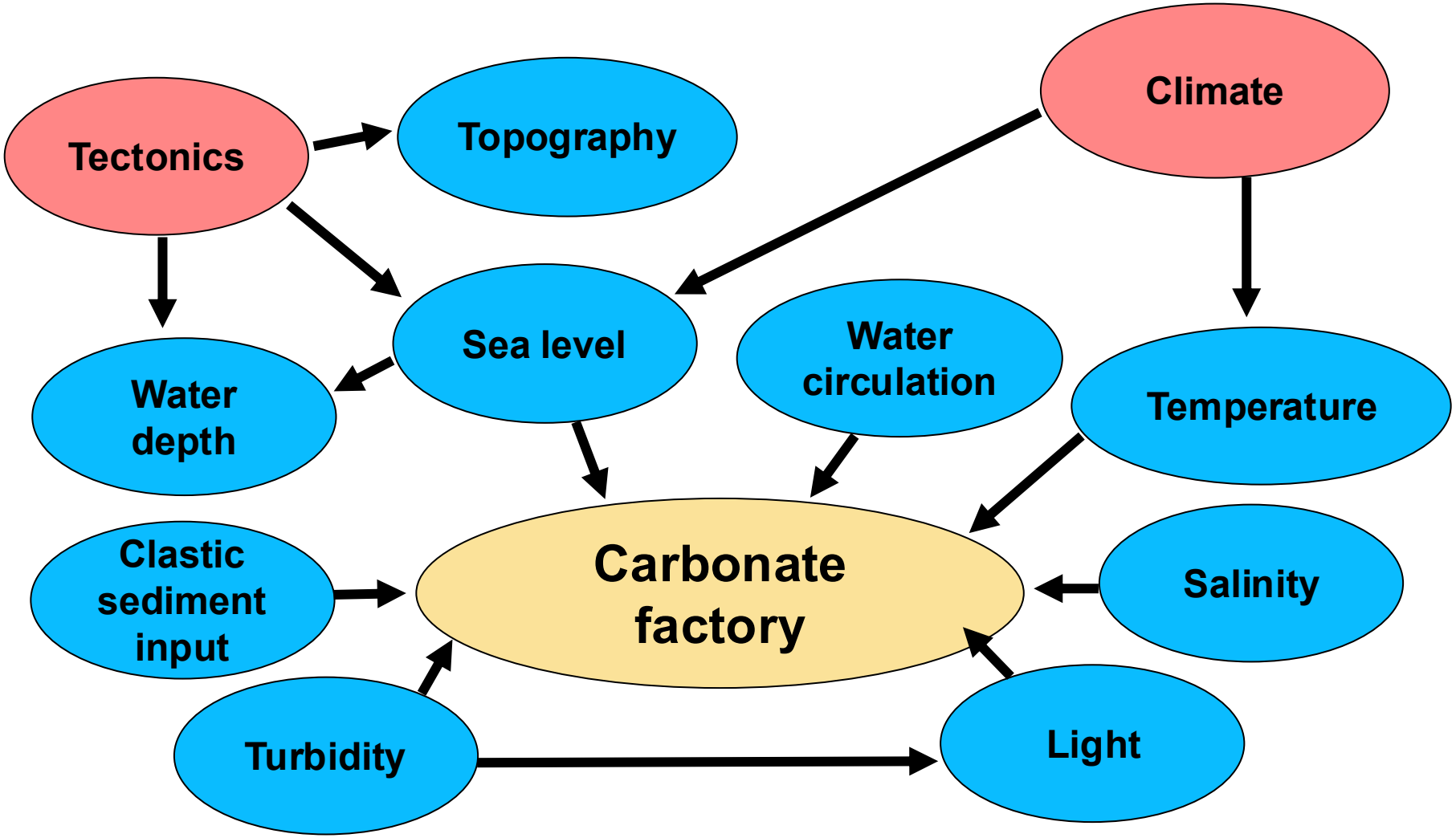


Andrieu (2016)

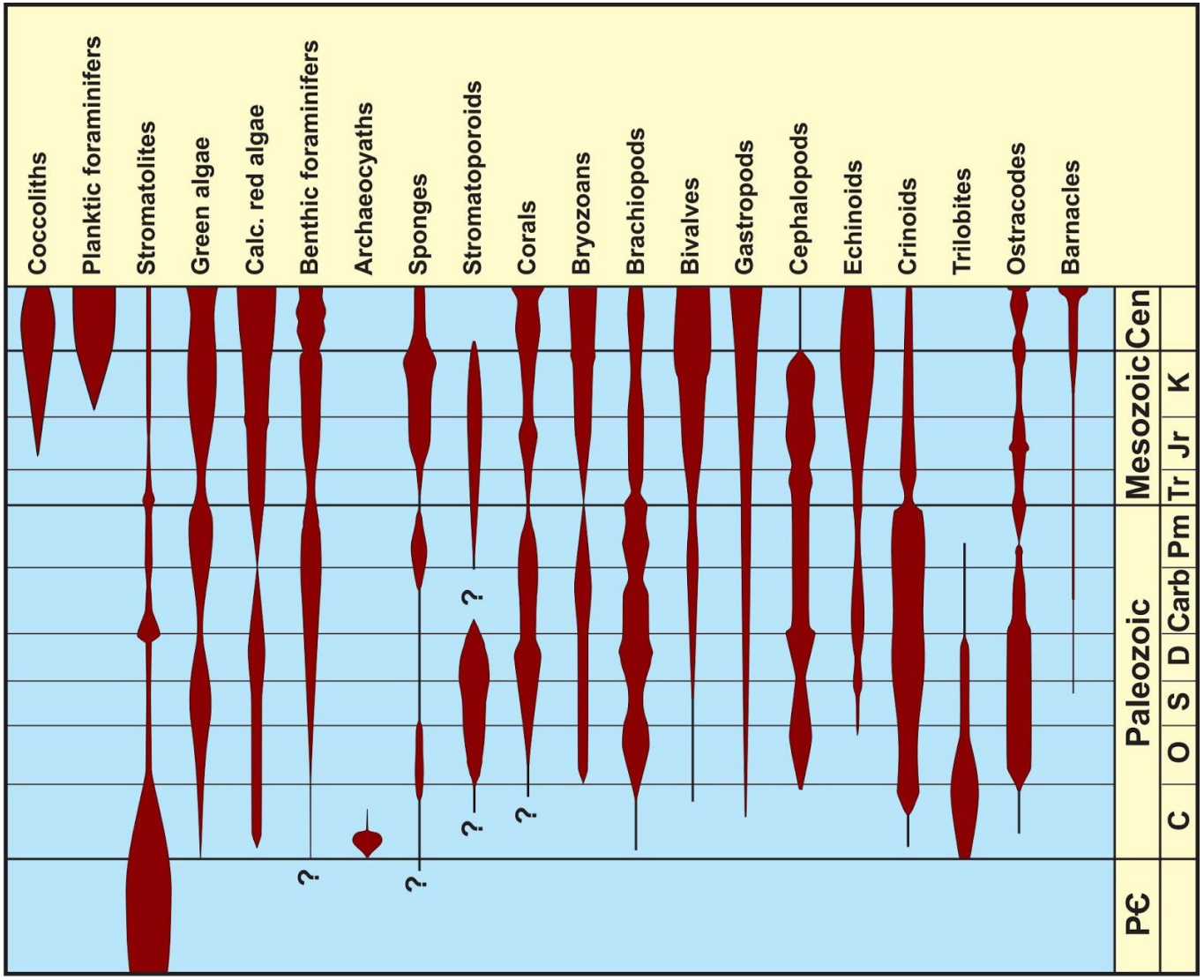
- **Callovian-Early Oxfordian: Wetter climate, increased siliciclastic input, transgressive anoxia do not favor carbonate productivity**
- **Middle-Late Oxfordian: Drier climate, low siliciclastic input favor carbonate productivity and build up development**

Carbonates through time

Carbonates through time
Controls on carbonate sedimentation

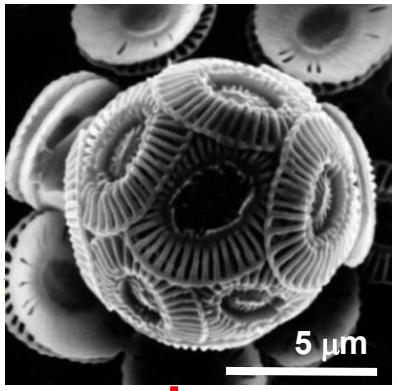


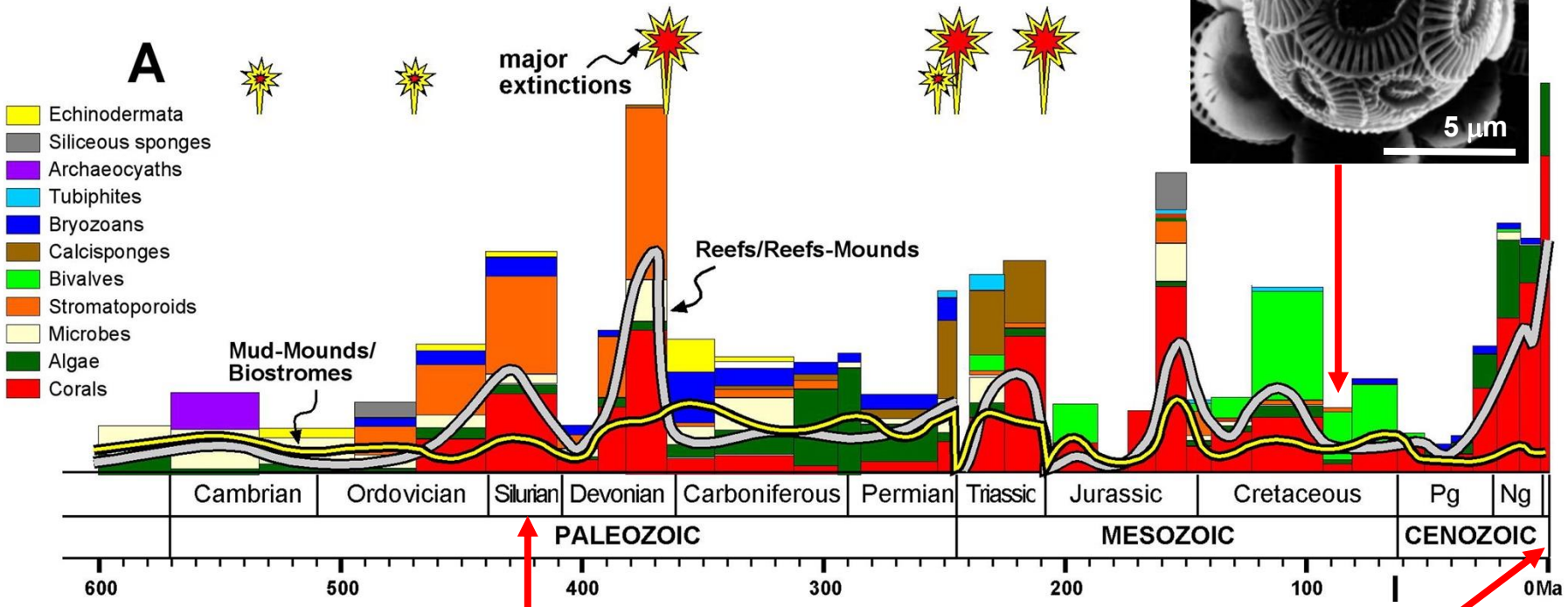
Carbonates through time
Influence of evolution



- **Changes in biota through time mean that uniformitarianism does not strictly apply**

Carbonates through time
Influence of evolution

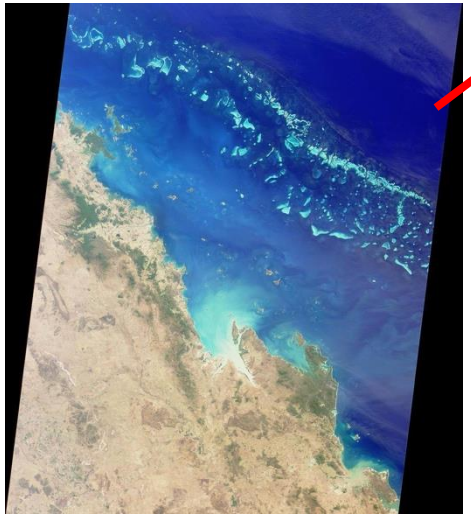




Pomar (2016)



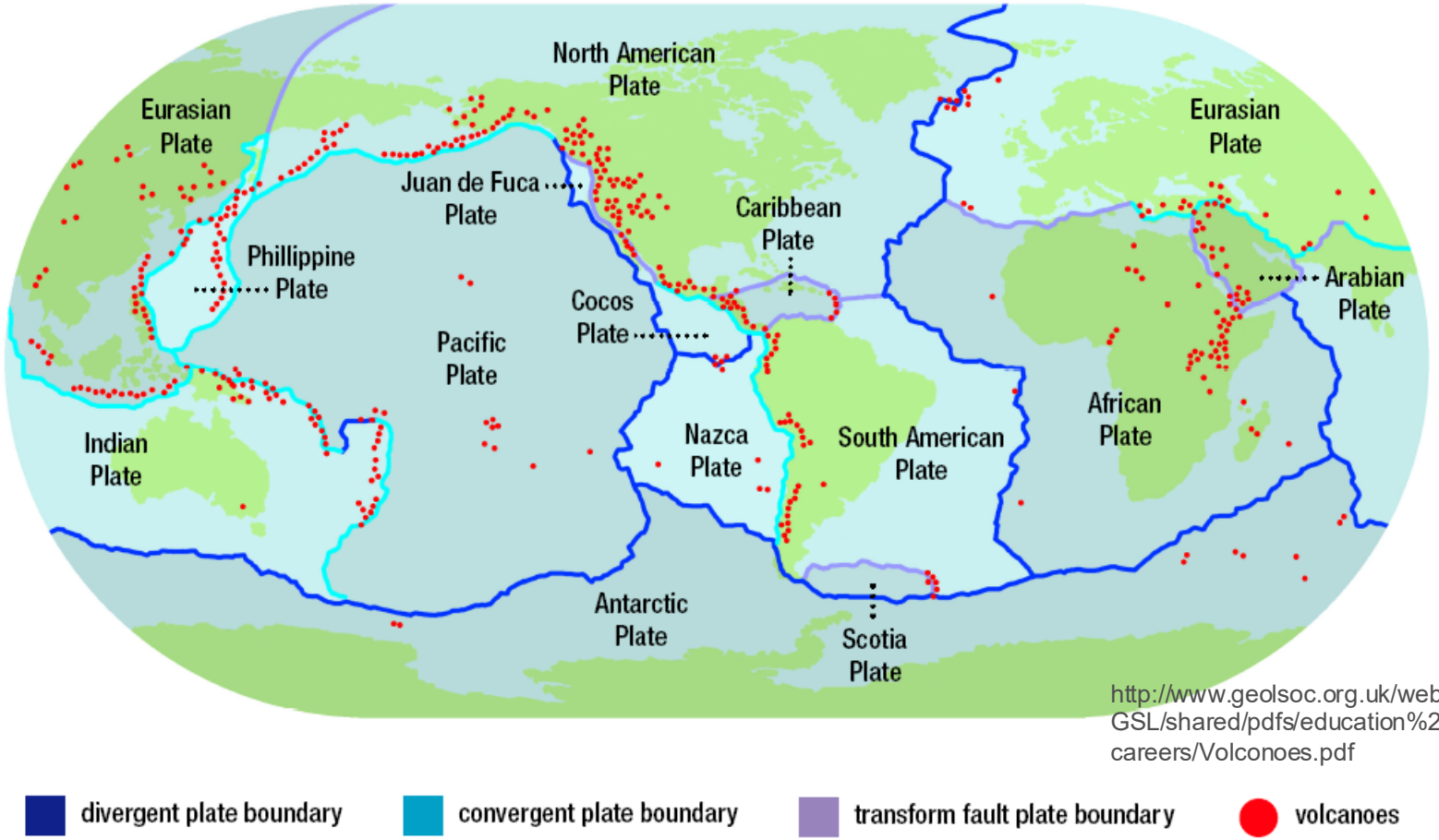
http://www.uni-graz.at/bernd.kaufmann/mudmounds.htm



© NASA

Carbonates through time

Plate tectonics

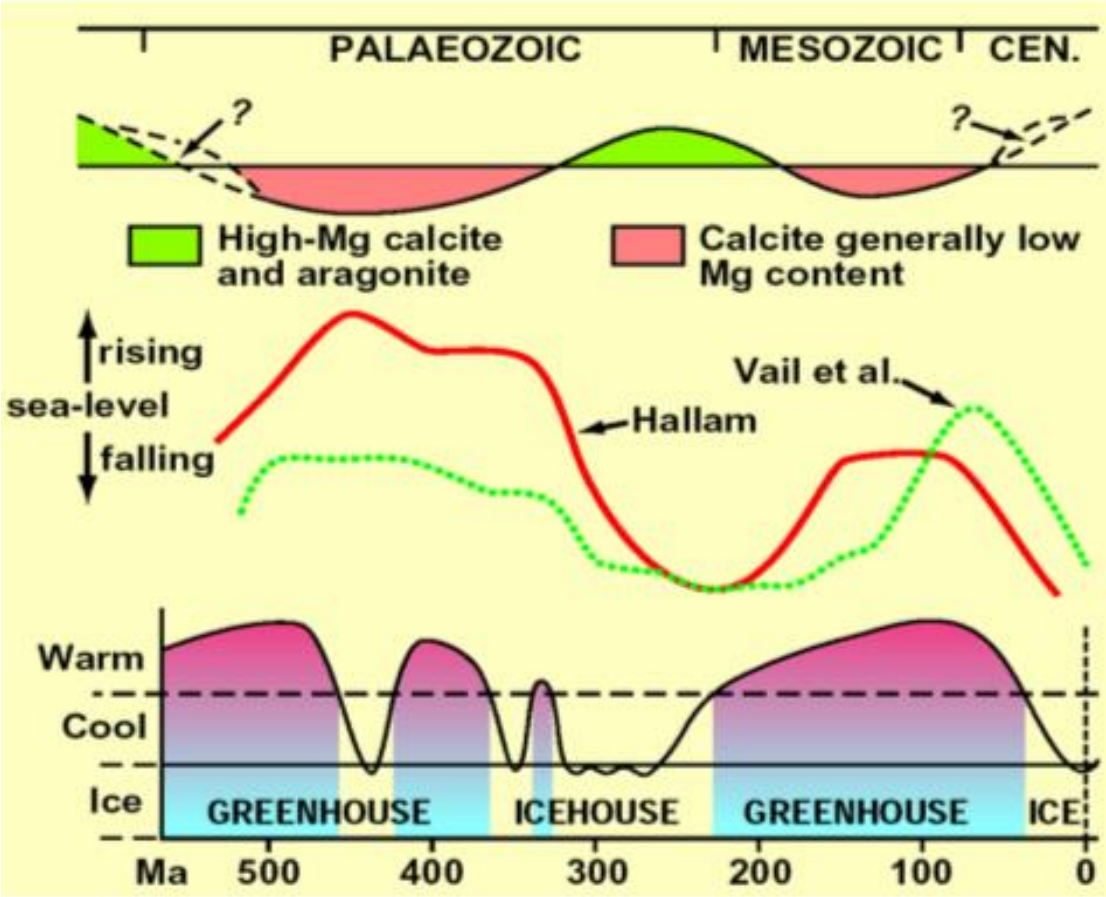


http://www.geolsoc.org.uk/webdav/site/GSL/shared/pdfs/education%20and%20careers/Volconoes.pdf

- Rates of sea floor spreading influences the abundance of atmospheric CO_2 (and thus precipitation/dissolution)
- Regional and local tectonism may control platform geometry
- Plate tectonics controls the distribution of clastic sediment: influencing hinterland topography and river drainage

Carbonates through time
Ocean chemistry

- Stratigraphic variation in ooid mineralogy (Sandberg, 1983)
- Recognition of ”aragonite seas” and “calcite seas” denoting dominant primary marine carbonate mineralogy
- Link to global sea level and greenhouse/icehouse cyclicity

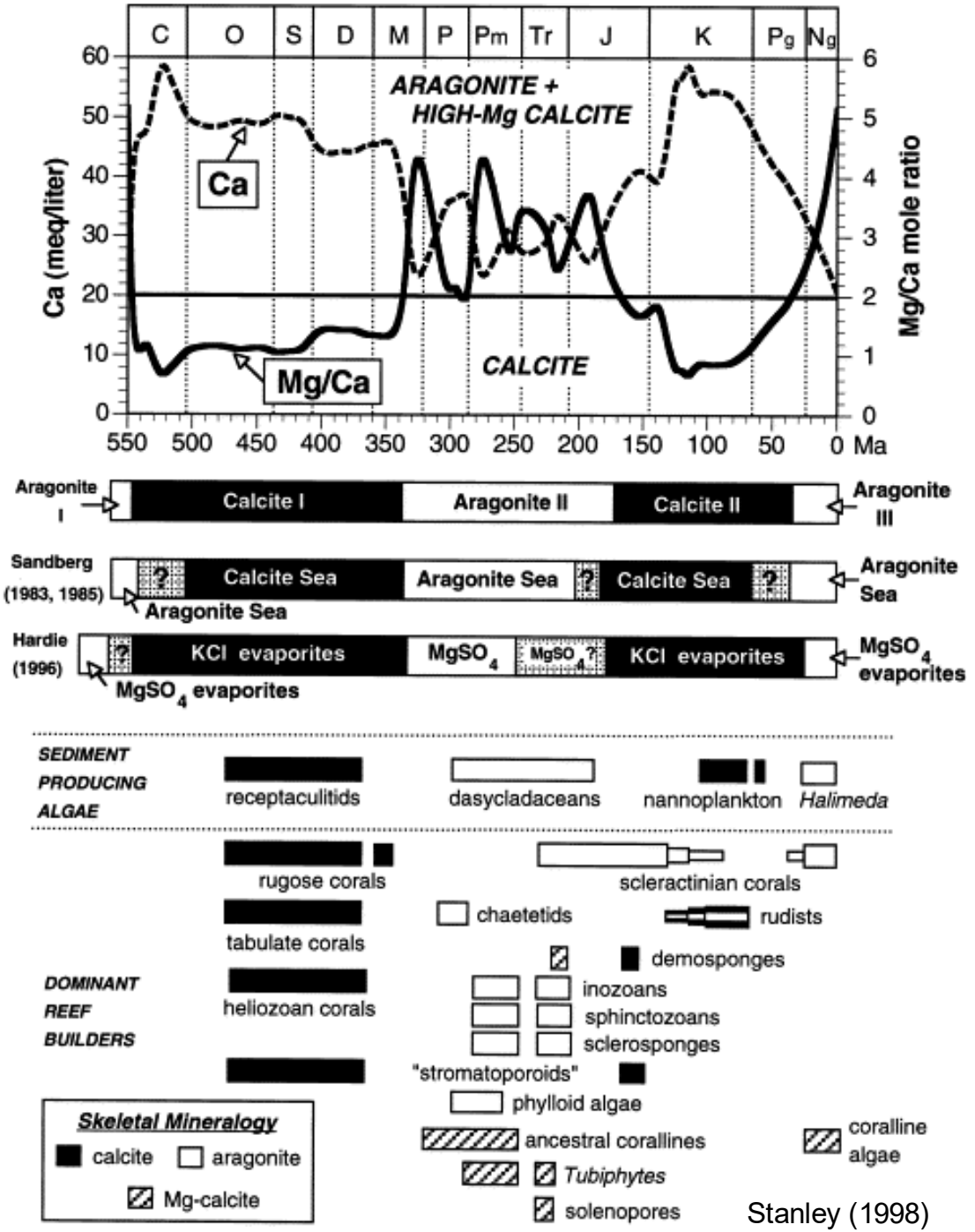


From Tucker (1993)

Carbonates through time

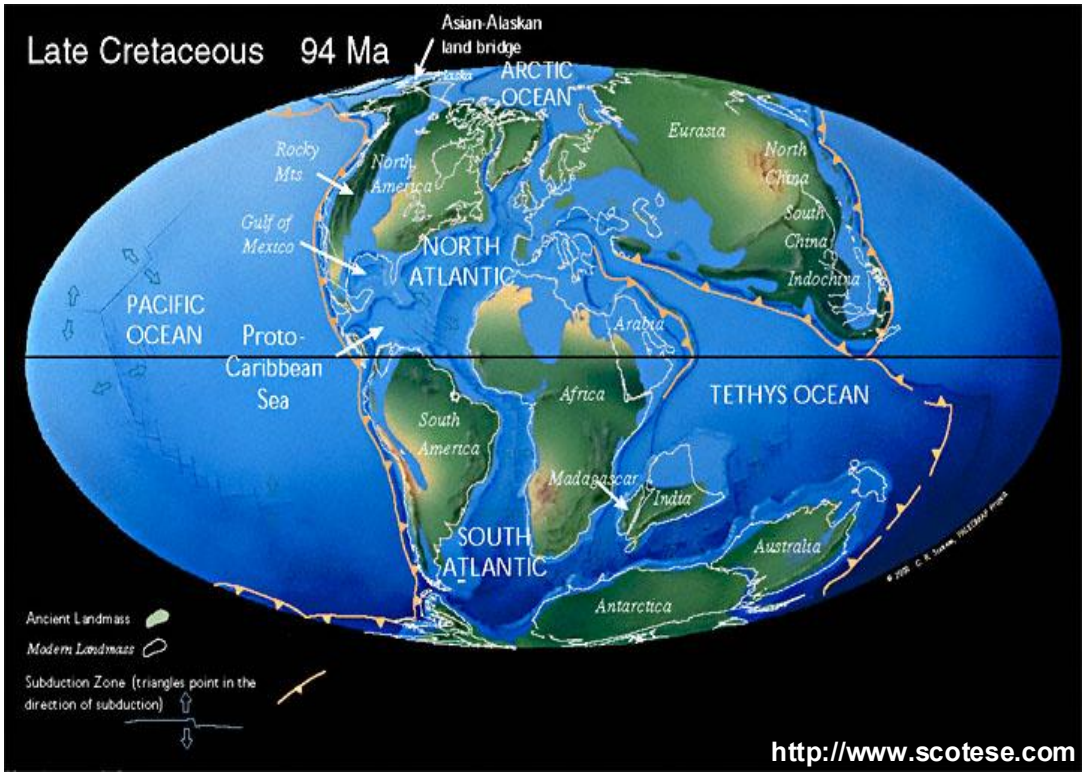
Ocean chemistry

- Precipitation of low magnesium calcite is favoured by high pCO_2 and low Mg/Ca ratios
- Precipitation of aragonite and high magnesium calcite is favoured by low pCO_2 and high Mg/Ca ratios (>5)



Stanley (1998)

Carbonates through time
Greenhouse

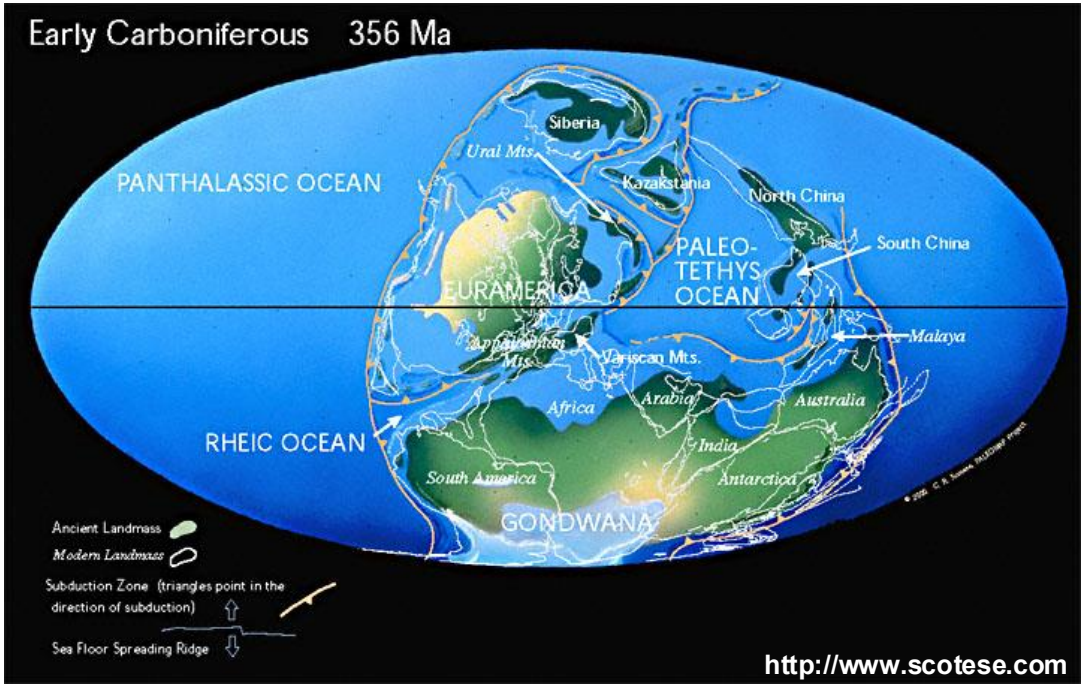


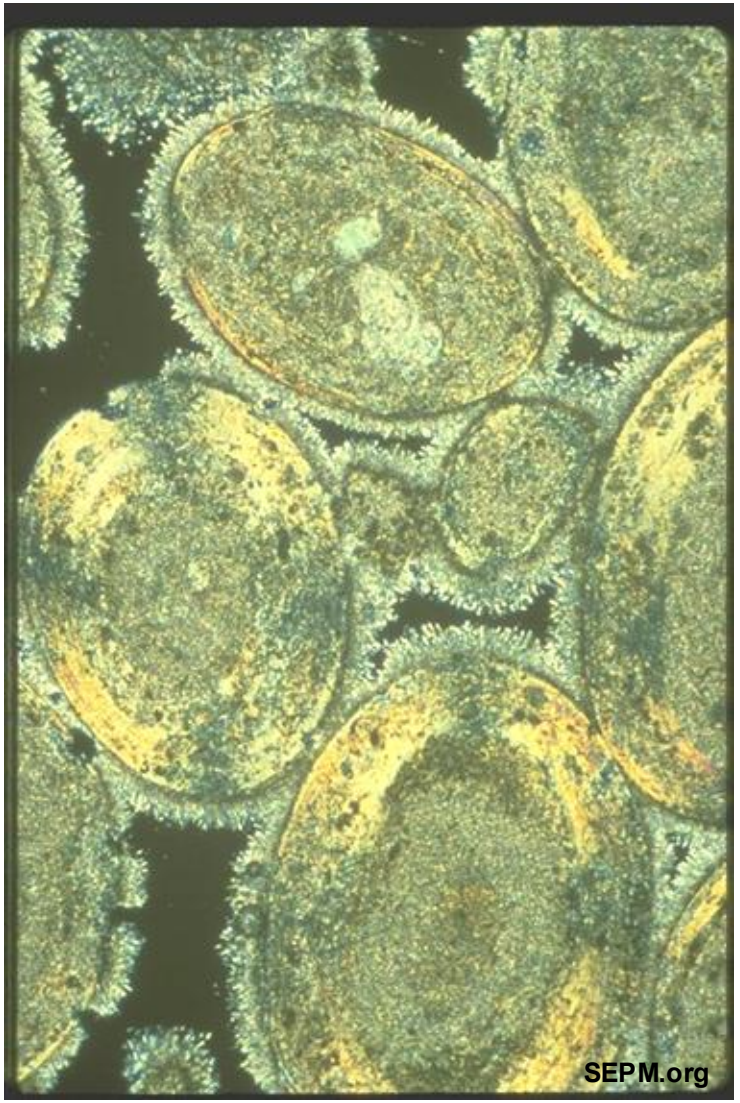


- High spreading rates
- Dispersed continents
- Relatively high sea level - large continental areas flooded
- Low latitudinal temperature gradient- warm at the poles
- No continental ice

- Increased pCO_2 by production of CO_2 from processes at subduction zones and MOR
- Lower Mg/Ca ratio of seawater via hydrothermal alteration of basalts at MOR
- Dominantly calcite seas

Carbonates through time
Icehouse





- Low spreading rates
- Assembled continents
- Relatively low sea level - little continental areas flooded
- High latitudinal temperature gradient - cold at the poles
- Continental ice sheets at the poles

- Decreased pCO_2 through increased photosynthesis
- Increased Mg/Ca ratio: increased Mg-supply via continental weathering, removal of Ca via evaporite precipitation
- Dominantly aragonitic seas

Carbonate dynamics
Summary

- Carbonate facies provide an excellent stratigraphic record of past global environmental changes through their sedimentology, fossil content and geochemistry.
- Facies prediction is fundamental to understand response of (bio)sedimentary systems to environmental change, and for applied geology (groundwater, petroleum, mining, quarrying….).
- Careful observations on fossil content, microfacies, and geochemistry of carbonates and associated sediments are key to facies analysis and prediction.
- Understanding dynamics of depositional systems and their response to relative sea level change is fundamental when predicting facies distribution and geometries in time and space.
- Carbonate response to relative sea level can be understood by comparing rates of carbonate production and sea level rise/fall.